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MSci Research Project

# Variational Methods for Option Pricing


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**Abstract**

Type your abstract here. The abstract is a summary of the contents of the project. It should be brief but informative, and should avoid technicalities as far as possible.

# Acknowledgments

## Plagiarism statement

The work contained in this thesis is my own work unless otherwise stated.

*Signature:* James Wu
*Date:* April 25, 2026

# Contents

# Chapter 1

# Introduction

## 1.1 Motivation

Options are derivative contracts that give the holder the right, but not the obligation, to buy or sell an underlying asset at a specified strike price $K$, specified at the time of option purchase. The act of buying or selling the underlying asset is called exercising the option, and is done at a predetermined expiry time $T$ (for European options) or at any time before expiration (for American options). A contract allowing the holder to buy the underlying asset is called a call option, and that allowing the holder to sell the underlying asset is called a put option.

These options are very important financial instruments, and tens of millions of contracts are traded every day just for the S&P 500 index alone, as they are useful for hedging and speculation. For that reason, it is important to be able to price these options accurately. This is a non-trivial task, as the price of an option depends on many factors, such as the underlying asset price which is generally modelled as a stochastic process.

## 1.2 Literature review

The pioneering work in option pricing was done by Black and Scholes [1], who derived a closed-form solution for the price of the European call option under the assumption that the underlying asset price follows a geometric Brownian motion. This model, known as the Black-Scholes model, is considered the foundational work in option pricing theory. However, it relies on several assumptions such as constant risk-free rate and volatility, which is not always realistic in practice. As a result, Heston [2] derived the European option prices of the proposed stochastic volatility model, allowing the volatility to vary over time to capture more facets of market dynamics.

The American option is much more difficult to price, since it allows the holder to exercise the option at any time before expiration, making it analogous to an optimal stopping problem that does not admit a closed-form solution in general. As a result, various numerical methods have been developed to tackle this problem, such as the binomial tree method [3, 4] for Black-Scholes and Heston respectively, Monte-Carlo simulation methods [5] and more recently variational methods where the optimal stopping problem is formulated as a variational inequality, solved by deep neural networks [6, 7]. This method is adapted to incoporate fractional Sobolev norms to guarantee convergence of the solution to the true option price instead of just pointwise convergence, which has been particularly developed for American option pricing [8].

## 1.3 Outline of project

Our project focuses on the Black-Scholes and Heston models, for which we will progressively build benchmarks for increasingly complex options. For each model, we begin by deriving the price of a European call option, which serves as a benchmark for the binomial tree method. The binomial tree easily extends to American options, providing a benchmark for the variational neural network method, which we implement with both the standard loss function and the Sobolev loss function. We then extend the analysis to multi-asset options. We first price options on the product of assets, since there exists an equivalent single-asset reduction and can be validated against the benchmarks already established. This foundation allows us to tackle payoffs without such a reduction, such as the maximum or minimum of several assets. Once the numerical results across these methods agree, we will apply our pricing scheme to two further objectives: computing the free boundary of the American option under varying market conditions, and evaluating the Greeks (the sensitivities of the option price to model parameters) to gain insight into exposures and hedging strategies.

# Chapter 2

# Mathematical Background

## 2.1 Preliminaries

We will first assemble the necessary mathematical background to understand the option pricing problem and build some foundations into the methods we will be using to solve it. The definitions and results are mostly standard, and can be found in textbooks such as [9, 10] but we will cover them here for self-containment and to set up notation for the rest of the project. As a result, we will only cover the main concepts and not delve too deeply into the details, since our focus is more on the numerical methods and their applications.

### 2.1.1 Probabilistic Setup

The pricing problem is inherently probabilistic, as we are uncertain about the future evolution of the underlying asset price, thus requiring a rigorous mathematical framework to model this uncertainty. The formalisation of the underlying randomness in our model for the market will be done through probability spaces, and the arrival of information in the market through time through filtrations.

**Definition 2.1.1** (Probability space). A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where $\Omega$ is the sample space, $\mathcal{F}$ is a $\sigma$-algebra on $\Omega$, and $\mathbb{P} : \mathcal{F} \to [0, 1]$ is a probability measure such that $\mathbb{P}(\Omega) = 1$.

**Definition 2.1.2** (Filtration, filtered probability space). A *filtration* on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a non-decreasing family $(\mathcal{F}_t)_{t \geq 0}$ of sub-$\sigma$-algebras of $\mathcal{F}$, that is to say, $\mathcal{F}_s \subseteq \mathcal{F}_t \subseteq \mathcal{F}$ for all $s \leq t$. The quadruple $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \geq 0}, \mathbb{P})$ is called a *filtered probability space*.

We will also define a Martingale, which formalises the idea that future expectations of a process are equal to the current value, which is a key concept in the risk-neutral pricing framework.

**Definition 2.1.3** (Martingale). A process $M = (M_t)_{t \geq 0}$ is a *martingale* with respect to the filtration $\{\mathcal{F}_t\}$ and the probability measure $\mathbb{P}$ if $\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s$ for all $0 \leq s \leq t$.

The Markov assumption is also commonly made in financial modelling, which states that the future evolution of the process only depends on the current state and not on the past history. It is useful because it greatly simplifies models, and hence we will employ it in our project as well.

**Definition 2.1.4** (Markov process). A process $X = (X_t)_{t \geq 0}$ is a *Markov process* with respect to the filtration $\{\mathcal{F}_t\}$ if

$$\mathbb{E}[f(X_t) \mid \mathcal{F}_s] = \mathbb{E}[f(X_t) \mid X_s] \tag{2.1}$$

for all $0 \leq s \leq t$ and all bounded measurable functions $f$.

### 2.1.2 Stochastic processes

The market in practice is not deterministic, so we will mathematically construct our model with stochastic processes, indexed by time in our application.

**Definition 2.1.5** (Stochastic process)**.** A *stochastic process* is a family of random variables $\{X_t : t \geq 0\}$ defined on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

More specifically, we will use Brownian motion to model the underlying asset price dynamics, which is a continuous-time stochastic process with continuous paths and stationary independent increments.

**Definition 2.1.6** (Brownian motion)**.** A real-valued process $W = (W_t)_{t \geq 0}$ on the probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a *Brownian motion* if it satisfies the following properties:

- $W_0 = 0$ almost surely
- The increment $W_t - W_s$ is normally distributed with mean $0$ and variance $t - s$ for all $0 \leq s < t$
- The increments $W_{t_1} - W_{t_0}, \ldots, W_{t_n} - W_{t_{n-1}}$ are independent for all $0 \leq t_0 < t_1 < \cdots < t_n$
- The sample paths $t \mapsto W_t(\omega)$ are continuous

The existence of this process is established by Wiener [11] and is very powerful in modelling the randomness in the financial market.

### 2.1.3 Risk-neutral pricing

We will now formalise the financial market and establish the principle of arbitrage-free valuation that will underpin our option pricing problem.

In our model, the market involves a fixed horizon $T > 0$ (our expiry) and a filtered probability space $(\Omega, \mathcal{F}, \{\mathcal{F}_t\}_{t \in [0,T]}, \mathbb{P})$ where $\{\mathcal{F}_t\}$ is the filtration generated by the underlying Brownian motion. The market consists of a risk-free asset $B_t$ that is deterministic and follows the evolution

$$B_t = \exp\left(\int_0^t r(s)\,\mathrm{d}s\right) \tag{2.2}$$

and one or more risky assets $S_t$ that follows an Itô process.

It is important that the market is arbitrage-free, which means that there does not exist any trading strategy that can generate a positive profit with zero initial investment and zero risk. This trading strategy $\phi$ must be self-financing, which means that that any change in the value of the portfolio is only due to changes in the assets held, without any external cash flow.

**Definition 2.1.7** (Self-financing trading strategy)**.** A *self-financing trading strategy* $\phi$ is a predictable process such that the value of the portfolio $V_t = \phi_t^B B_t + \phi_t^S S_t$ satisfies

$$\mathrm{d}V_t = \phi_t^B \mathrm{d}B_t + \phi_t^S \mathrm{d}S_t \tag{2.3}$$

**Definition 2.1.8** (Arbitrage)**.** An *arbitrage* is some trading strategy $\phi$ such that $V_0 = 0$, $V_T \geq 0$ almost surely and $\mathbb{P}(V_T > 0) > 0$.

By the fundamental theorem of asset pricing, the market is arbitrage-free if and only if there exists a measure $\mathbb{Q}$ such that $\mathbb{Q}$ is equivalent to $\mathbb{P}$ (they have the same null sets) and the discounted price process $\tilde{S}_t = \frac{S_t}{B_t}$ is a martingale under $\mathbb{Q}$. We will assume that our market is complete (any derivative can be replicated), and hence such a measure $\mathbb{Q}$ will be unique.

Given that the market is arbitrage-free, we can price any derivative with payoff $g(\cdot)$ at time $t$ by taking the discounted expectation of the payoff under the risk-neutral measure $\mathbb{Q}$, giving us the risk-neutral pricing formula for European derivative with maturity $T$:

$$v(t, S) = B_t \mathbb{E}^{\mathbb{Q}} \left[ \frac{g(S_T)}{B_T} \,\middle|\, \mathcal{F}_t \right] \tag{2.4}$$

Throughout this project specifically, we will deal with constant interest rates, so the bond price process reduces to $B_t = e^{rt}$ where $r$ is the constant risk-free rate. Additionally, we will also assume the Markov property (2.1.4). The arbitrage-free price of the European derivative will therefore be given by

$$v(t, S) = \mathbb{E}^{\mathbb{Q}} \left[ e^{-r(T-t)} g(S_T) \,\middle|\, S_t = S \right] \tag{2.5}$$

### 2.1.4 Optimal stopping problem and free boundary

The American option gives the holder the right to exercise the option at any time before maturity, which thus gives us the interpretation of the option price as the solution to an optimal stopping problem, as we need to additionally identify when we should exercise the option.

**Definition 2.1.9** (Stopping time). A random variable $\tau : \Omega \to [0, \infty]$ is a *stopping time* if $\{\tau \leq t\} \in \mathcal{F}_t$ for all $t \geq 0$.

The price of the American option can thus be expressed as the $\mathbb{Q}$-expectation of the discounted payoff at the optimal stopping time $\tau^\star$ that maximises the expected payoff, giving the optimal stopping problem

$$v(t, S) = \sup_{\tau \in [t,T]} \mathbb{E}^{\mathbb{Q}} \left[ e^{-r(\tau-t)} g(S_\tau) \,\middle|\, S_t = S \right] \tag{2.6}$$

which is simply (2.5) with the additional optimisation over the stopping time $\tau$.

This optimal $\tau^\star$ is the first time that the option holder chooses to exercise the option, which motivates the notion of a free boundary, which is the boundary separating separating the exercise region and the continuation region.

Let's work with the case $t = 0$ for simplicity, but the same logic applies for any $t \in [0, T)$. For the American put, the optimal stopping problem (2.6) can be rewritten as

$$v(0, S) = \sup_{\tau \in [0,T]} \mathbb{E}\left[ e^{-r\tau} (K - S_\tau)^+ \mid S_0 = S \right] \tag{2.7}$$

which is achieved by an optimal stopping time $\tau^\star$ taking the form

$$\tau^\star = \inf\{\tau \geq 0 : S_\tau \leq b^\star(\tau)\} \tag{2.8}$$

for some optimal exercise boundary $b^\star(\tau)$ [5]. This is illustrated in Figure 2.1, where we simulate one realisation of the asset price assuming a geometric Brownian motion and plot it against the exercise boundary. The illustration is based on the binomial tree solution on the American put option, which has the boundary

$$b^\star(\tau) = \sup\{S : c(\tau, S) = g(S)\} \tag{2.9}$$

which is found using binary search here, and $c(t, S)$ is the continuation value of the option with payoff $g(S)$.

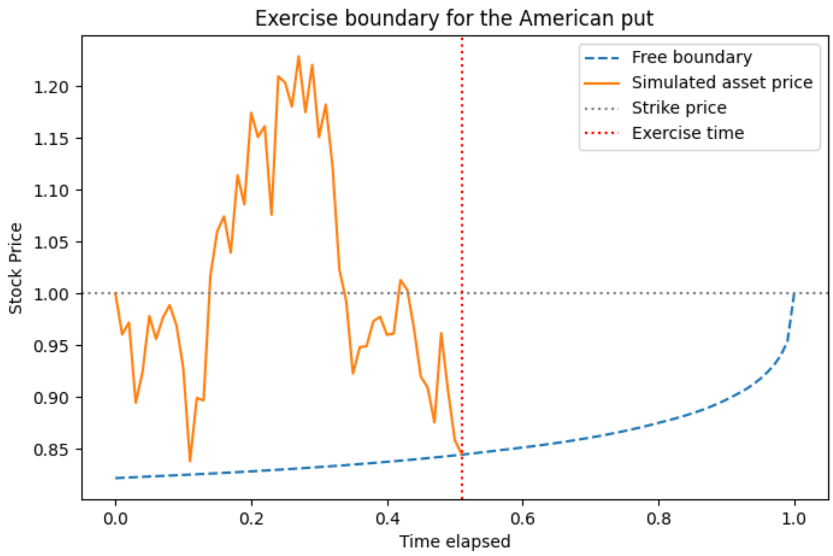


**Figure 2.1** Exercise boundary for the American put (adapted from [5])

### Soft classification of the free boundary

This formulation describes a hard classification problem, where we classify points in the time-stock price space binarily to be either in the continuation region or the exercise region. However, our numerical solutions to the pricing problem are not exact and there is some level of error in the solution that comes from the Monte Carlo simulations upon the space, the numerical differentiation used in training and the early stopping in the training process. Therefore, it would be more intuitive to perform a soft classification instead, where we assign probabilities to points in the time-stock price space to be in the continuation region or the exercise region, which would give us a more intuitive visualisation of the free boundary and how it evolves over time. We will do this by extending hard classification: that is to say, the continuation region $A$ is

$$A = \{(t,S) : c(t,S) > g(S)\} \cup \{(t,S) : g(S) = 0\} \tag{2.10}$$

where intuitively, the first set is those where the continuation value is greater than the early exercise value (and hence it is better to continue), and the second set is those where the payoff is zero, which are at-the-money and out-of-the-money options that are worthless under immediate exercise.

### Sigmoid soft classification

We will consider two methods to extend the definition of the continuation region to a soft classification. The first one is to use a sigmoid function $s : \mathbb{R} \to (0,1)$ defined as

$$s(x) = \frac{1}{1+e^{-x}} \tag{2.11}$$

to smoothly transition the continuation probability from $0$ to $1$ as the value function transitions from being below the payoff to being above the payoff. As the sigmoid function is the CDF of the logistic distribution, we can interpret this as modelling the noise in the value function as a logistic distribution. The effect of the noise can be controlled by introducing the softness parameter $\tau > 0$, and our continuation probability function can be defined on the relative difference from the payoff as

$$P_{\text{cont}}(d) = s\left(\frac{d}{\tau}\right), \qquad d := \frac{c(t,S) - g(S)}{g(S)} \tag{2.12}$$

To tune the softness parameter, we can define a threshold $I \in (0.5, 1)$ such that we want the continuation probability to be $I$ at some relative difference $\epsilon > 0$, i.e. $P_{\text{cont}}(\epsilon) = I$, which we could solve with some simple algebraic manipulation to get

$$\tau = \frac{\epsilon}{\ln\left(\frac{I}{1-I}\right)} \tag{2.13}$$

**Gaussian noise soft classification**

The second method is to assume Gaussianity of the noise in our estimate $\hat{c}$ of the continuation value, and that our approximated price is unbiased. In other words, we have

$$\hat{c}(t,S) = c(t,S) + \epsilon, \qquad \epsilon \sim \mathcal{N}(0, \tilde{\sigma}^2) \tag{2.14}$$

where $c$ is the true (latent) continuation value and $\tilde{\sigma}$ is the standard deviation of the solution noise, which is also unknown. We can then perform solve the problem $n$ times to get the different approximated continuation values $\hat{c}_1, \ldots, \hat{c}_n$. This allows us to use the sample mean $\bar{c}$ as our unbiased estimate of the continuation value function paired with the sample standard deviation $\hat{\sigma}$ as our estimate. Then $\bar{c}$ is distributed according to a $t$-distribution with $n-1$ degrees of freedom, and we can compute the continuation probability as

$$P_{\text{cont}}(t,S) = \mathbb{P}_{t_{n-1}}\left(\frac{\bar{c}(t,S) - g(S)}{\hat{\sigma}(t,S)/\sqrt{n}} > 0\right) \tag{2.15}$$

which is simple as the CDF of the $t$-distribution is readily available.

The caveat with this method is that it requires multiple simulations of the solution, which becomes infeasible for more complex models and higher dimensions, but it gives us a more statistically grounded way to compute the continuation probability.

All that is left is to find a method to obtain the continuation value function, as it is generally latent and not directly observable, given that we predict the American option price that already takes the maximum of the continuation value and the payoff. However, we can use

$$c(t,S) = \mathbb{E}\left[e^{-rh} f(t+h, S_{t+h}) \,\middle|\, S_t = S\right] \tag{2.16}$$

where $h > 0$ is a small time step and $f$ is the American option price function. We can obtain an unbiased estimation of this expectation by Monte Carlo: we simulate the asset price $S_{t+h}$ at time $t+h$ given $S_t = S$ many times and evaluate the option price function for each simulation, then taking the mean.

### 2.1.5 Monte Carlo integration

Monte Carlo is a powerful way to estimate integrals that may be intractable or not have a closed-form solution, which will be useful through our project as expectations can be expressed as integrals under the probability measure. The idea is to use the law of large numbers to estimate the integral by sampling from some distribution and evaluating the integrand at the sampled points.

Suppose we want to compute the integral of some function $\phi$ over a domain $\Omega$:

$$I = \int_{\Omega} \phi(x) \, \mathrm{d}x \tag{2.17}$$

If we are able to sample iid from some distribution with probability density function $q$ that has positive density on $\Omega$, then we can rewrite the integral as

$$I = \int_{\Omega} \frac{\phi(x)}{q(x)} q(x) \, \mathrm{d}x = \mathbb{E}\left[\frac{\phi(X)}{q(X)}\right] \tag{2.18}$$

where the expectation is taken under the distribution with density $q$. This expectation can be estimated by sampling $X_1, \ldots, X_N$ iid from $q$ and taking the sample mean:

$$\hat{I} = \frac{1}{N} \sum_{j=1}^{N} \frac{\phi(X_j)}{q(X_j)} \tag{2.19}$$

This is known as importance sampling [12], and our choice of $q$ can affect the variance of the estimator $\hat{I}$. In practice, we can choose $q$ to be uniform over $\Omega$, giving us the simple estimator

$$\hat{I} = \frac{1}{N} \sum_{j=1}^{N} \phi(X_j) \tag{2.20}$$

## 2.2 State Space Discretisation

### 2.2.1 Background

Recall that the option value function $v(t, S)$ is the solution to the optimal stopping problem

$$v(t, S) = \sup_{\tau \in [t,T]} \mathbb{E}\left[e^{-r(\tau - t)} g(S_\tau) \mid S_t = S\right] \tag{2.21}$$

There does not exist a closed-form solution for this problem in general, and thus we have to resort to numerical methods. We first consider methods that discretise the continuous time into a finite grid $\{t_i\}_{i=1}^{N}$ where we have

$$0 = t_0 < t_1 < \cdots < t_N = T \tag{2.22}$$

which we can use to index the underlying state variable $S_{t_i}$ at time $t_i$. Note that the state variable $S_{t_i}$ could depend on multiple variables, such as in the Heston model, where we have a stochastic variance term encoded in the asset price. The goal is to compute the arbitrage-free price $v(0, S_0)$ at time $t = 0$ given the initial state $S_0$. As we know the solution must satisfy the terminal condition $v(T, S) = g(S)$ for all $S$, we can use backward induction to compute the option value at each time step. This gives rise to the class of numerical methods known as tree methods.

### 2.2.2 Tree Methods

In a tree model, we discretise the time-state space into a finite grid, where the state variable $S_{t_i}$ can take a finite number of values at each time step given the previous one. We will particularly explore binomial trees in this project, where each state can evolve to one of two possible states at the next time step. Formally, suppose at time $t_n$ we have the state $S_{t_n} \in \{S_n^1, \dots, S_n^{M_n}\}$ where $M_n$ is the number of possible states at time $t_n$. Then at time $t_{n+1}$, each state $S_n^i$ can evolve to the state $S_{n+1}^j$ with transition probability

$$p_{ij}^{(n)} = \mathbb{P}(S_{n+1} = S_{n+1}^j \mid S_n = S_n^i) \tag{2.23}$$

This is a discrete-time Markov chain, and to identify the transition probabilities, we can use the risk-neutral condition which states that the discounted price process must be a martingale under the risk-neutral measure. This gives us the condition

$$\sum_{j=1}^{M_{n+1}} p_{ij}^{(n)} S_{n+1}^j = e^{r\Delta t} S_n^i \tag{2.24}$$

where $\Delta t = t_{n+1} - t_n$ and $r$ is the risk-free rate.

With these transition probabilities, we can compute the option value at time $t_n$ given the option value at time $t_{n+1}$ by using backwards induction, which gives us the recursive relation

$$v(t_n, S_n^i) = \max\left(g(S_n^i), e^{-r\Delta t} \sum_{j=1}^{M_{n+1}} p_{ij}^{(n)} v(t_{n+1}, S_{n+1}^j)\right) \tag{2.25}$$

where we take the maximum of the payoff and the discounted expected future price to account for early exercise for the American option. This maximum operator can easily be lifted for all time steps for European options, or only imposed on certain time steps for Bermudan options, leaving the tree method a very flexible method. This backwards induction is applied iteratively starting from the terminal condition at time $t_N = T$ to compute the option value at time $t_0 = 0$.

An issue with tree methods is that the number of states $M_n$ can grow exponentially with the number of time steps $N$, leading to a very computationally expensive algorithm. To mitigate this, we use recombining trees where the states at each time step are designed to recombine with each other, so that the number of states $M_n$ grows linearly. Another issue is that, while flexible, the tree method can be difficult to implement for more complex models where the state depends on multiple variables or if we wish to extend it to options operating on multiple assets, but its reliability and interpretability make it a good benchmark for simpler models.

## 2.3 Physics-Informed Neural Networks

### 2.3.1 Motivation

Our option pricing problem can alternatively be formulated as a variational inequality which takes the form of a partial differential equation that is independent from the payoff $g(\cdot)$. These PDEs can be of high dimension, and are thus difficult to solve using traditional numerical methods due to the curse of dimensionality. Traditional methods such as finite difference methods require discretisation of the domain and the number of grid points (and hence algorithm complexity) scales exponentially $\mathcal{O}(N^d)$ with the dimension $d$ of the problem.

We can attempt a neural network solution to the PDEs, as neural networks are known to be universal approximators. That is, suppose we have the class of functions $\mathcal{F}$ over a compact set $S$ that can be represented by a neural network with a given architecture, with a continuous target function $f^\star$ that we would like to approximate. Then for any target accuracy $\epsilon > 0$, there exists a function $f \in \mathcal{F}$ such that

$$\sup_{x \in S} |f(x) - f^\star(x)| \leq \epsilon \tag{2.26}$$

If we have a squashing activation function $\Psi(\cdot)$ that is non-decreasing and satisfies the limits

$$\Psi(-\infty) = 0, \quad \Psi(\infty) = 1 \tag{2.27}$$

then the class of functions $\mathcal{F}$ that can be represented by a neural network with one hidden layer and activation function $\Psi$ is dense in the space of continuous functions on $S$ and thus is a universal approximator [13].

Since the option price function is continuous in the time-asset price domain, we can use a neural network to find an approximation. This is favourable because the time complexity scales with the number of parameters, which is not exponential. To do this, we will use the framework of physics-informed neural networks (PINNs) to solve the PDEs, which have been shown to be effective for solving high-dimensional PDEs [6], by avoiding mesh construction and leveraging the interpolation capability of neural networks [14]. The loss function of the neural network is designed to penalise deviations from the PDE as well as deviations from the boundary conditions. By training the neural network to minimise this loss function, we can obtain an approximation to the solution of the PDE. This method has been applied to option pricing problems in the literature, such as in [15] for the Black-Scholes model, but we will look to apply it to more complex models such as the Heston model as well as to options on multiple assets.

### 2.3.2 Methodology

We first provide a general setup for PINNs, before modifying it to tailor our specific problem of option pricing. Formally, suppose we have an unknown latent solution $u(t, x)$ which is governed by a PDE of the form

$$\frac{\partial u}{\partial t} + \mathcal{L}[u] = 0, \qquad t \in [0, T], x \in \Omega \tag{2.28}$$

where $\mathcal{L}$ is a (not necessarily linear) differential operator. Suppose additionally that this PDE is subject to the boundary conditions

$$u(0, x) = g(x), \qquad x \in \Omega \tag{2.29}$$

$$\mathcal{B}[u] = 0, \qquad t \in [0, T], x \in \partial\Omega \tag{2.30}$$

where $\mathcal{B}$ is a boundary operator. Then the unknown solution $u$ can be approximated by a deep neural network $f_\theta(t, x)$ where $\theta$ are the parameters of the neural network, trained by minimising a loss function composed of deviations from the PDE residual, boundary conditions and initial conditions [16]:

$$L(\theta) = \lambda_r L_r(\theta) + \lambda_{bc} L_{bc}(\theta) + \lambda_{ic} L_{ic}(\theta) \tag{2.31}$$

where the loss components are mathematically defined as

$$L_r(\theta) = \frac{1}{N_r} \sum_{i=1}^{N_r} \left| \mathcal{R}_\theta(t_r^i, x_r^i) \right|^2 \tag{2.32}$$

$$L_{bc}(\theta) = \frac{1}{N_{bc}} \sum_{i=1}^{N_{bc}} \left| \mathcal{B}[f_\theta](t_{bc}^i, x_{bc}^i) \right|^2 \tag{2.33}$$

$$L_{ic}(\theta) = \frac{1}{N_{ic}} \sum_{i=1}^{N_{ic}} \left| f_\theta(0, x_{ic}^i) - g(x_{ic}^i) \right|^2 \tag{2.34}$$

where the PDE residual $\mathcal{R}_\theta$ is defined as

$$\mathcal{R}_\theta(t, x) = \frac{\partial f_\theta}{\partial t}(t, x) + \mathcal{L}[f_\theta](t, x) \tag{2.35}$$

We additionally use the sampled points $\{x_{ic}^i\}_{i=1}^{N_{ic}}$ from the initial condition domain, $\{(t_{bc}^i, x_{bc}^i)\}_{i=1}^{N_{bc}}$ from the boundary condition domain and $\{(t_r^i, x_r^i)\}_{i=1}^{N_r}$ from the PDE domain, sampled uniformly in their respective domains at each iteration of the gradient descent algorithm.

There are different methods to choose the loss weights $\lambda_r, \lambda_{bc}, \lambda_{ic}$, such as using fixed weights or using adaptive weights that change during training, which we will discuss in Section 2.3.3.

For our specific problem of option pricing, we can modify the above setup to fit our variational inequality formulation. The PDE to solve is dependent on the model for the asset dynamics, and the boundary conditions are given by the payoff function of the option. In general, we will have to solve a problem of the form

$$\min\left\{-\frac{\partial u}{\partial t} + \mathcal{L}[u], u - g\right\} = 0, \qquad t \in [0,T], x \in \Omega \tag{2.36}$$

$$u(T,x) = g(x), \qquad x \in \Omega \tag{2.37}$$

$$\mathcal{B}_k[u] = 0 \qquad t \in [0,T], x \in \partial\Omega, k = 1, \ldots, K \tag{2.38}$$

where $k$ indexes the $K$ different boundary conditions. This is very similar to the general setup for PINNs, except that we have a variational inequality instead of a PDE, and we have a terminal condition instead of an initial condition. However, we can still use the framework by noting we can make the substitution $t \mapsto T-t$ to un-negate the negative partial derivative in the PDE term and to convert the terminal condition to an initial condition. The variational inequality can be handled by modifying the PDE residual loss (2.35) to

$$\mathcal{R}_\theta(t,x) = \min\left\{\frac{\partial f_\theta}{\partial t}(t,x) + \mathcal{L}[f_\theta](t,x), f_\theta(t,x) - g(x)\right\} \tag{2.39}$$

With everything in place, we can train the neural network with gradient descent:

$$\theta_{n+1} = \theta_n - \eta_n \nabla_\theta L(\theta_n) \tag{2.40}$$

where $\eta_n$ is the learning rate at iteration $n$, either kept constant or decayed over time with a schedule.

### 2.3.3 Adaptive Loss Weighting

While keeping the loss weights $\lambda_r, \lambda_{bc}, \lambda_{ic}$ fixed is simple, it can be difficult to tune correctly and can lead to suboptimal performance. Ideally, we would like to balance the different loss components so that they are of the same order of magnitude, which can be achieved by using adaptive loss weighting methods, as proposed in [16]. The idea is to evaluate the contribution of each loss component on the total loss and adjust the weights accordingly to balance their contributions. In practice, we will use a moving average to prevent the weights from changing too rapidly, leading to instability in training. Mathematically, we compute the adaptive weights

$$\hat{\lambda}_r = \frac{\|\nabla_\theta L_r(\theta)\| + \|\nabla_\theta L_{bc}(\theta)\| + \|\nabla_\theta L_{ic}(\theta)\|}{\|\nabla_\theta L_r(\theta)\|}$$
$$\hat{\lambda}_{bc} = \frac{\|\nabla_\theta L_r(\theta)\| + \|\nabla_\theta L_{bc}(\theta)\| + \|\nabla_\theta L_{ic}(\theta)\|}{\|\nabla_\theta L_{bc}(\theta)\|}$$
$$\hat{\lambda}_{ic} = \frac{\|\nabla_\theta L_r(\theta)\| + \|\nabla_\theta L_{bc}(\theta)\| + \|\nabla_\theta L_{ic}(\theta)\|}{\|\nabla_\theta L_{ic}(\theta)\|}$$

and update the global weights $\lambda = (\lambda_r, \lambda_{bc}, \lambda_{ic})$ with a smoothness $\alpha \in (0,1)$ as a moving average

$$\lambda_{\text{new}} = \alpha\hat{\lambda}_{\text{old}} + (1-\alpha)\hat{\lambda}_{\text{new}} \tag{2.41}$$

This is not performed at each iteration, but rather at a fixed frequency to prevent the weights from changing too rapidly. The literature [16] suggests to update the weights every $1000$ iterations with $\alpha = 0.9$ in general.

## 2.4 Sobolev Deep Neural Networks

We can explore a different method to price American puts with neural networks, where this time we work with fractional Sobolev norms to construct our loss function. This is motivated by the fact that we would be able to guarantee that our neural network converges to the true solution provided that its loss converges instead of just pointwise convergence. Additionally, the loss function components no longer depend on the payoff and is applicable for all situations, a desirable property for a general method.

The theory of Sobolev spaces are rich and we will not attempt to cover all the details to not divert the attention from the main focus of the project, but we will cover the necessary background to understand the method.

### 2.4.1 Sobolev Spaces

We first introduce the concept of Sobolev spaces. To begin, we consider those of integer order, defined over an arbitrary domain $\Omega \subset \mathbb{R}^n$, but we will generalise to fractional orders later. These are vector subspaces of various Lebesgue spaces $L^p(\Omega)$, which is the class of all measurable functions $u$ for which

$$\int_\Omega |u(x)|^p \, \mathrm{d}x < \infty \tag{2.42}$$

The Sobolev norms are functionals $\|\cdot\|_{m,p}$ where $m \in \mathbb{N}_+$ and $1 \leq p \leq \infty$, defined as follows:

$$\|u\|_{m,p} = \left( \sum_{0\leq|\alpha|\leq m} \|D^\alpha u\|_p^p \right)^{1/p} \qquad 1 \leq p < \infty \tag{2.43}$$

$$\|u\|_{m,\infty} = \max_{0\leq|\alpha|\leq m} \|D^\alpha u\|_\infty \tag{2.44}$$

where $\|\cdot\|_p$ is the norm in $L^p(\Omega)$. The Sobolev spaces are consequently those on which $\|\cdot\|_{m,p}$ is a norm [17]:

- $H^{m,p}(\Omega) \equiv$ the completion of $\{u \in C^m(\Omega) : \|u\|_{m,p} < \infty\}$ with respect to the norm $\|\cdot\|_{m,p}$
- $W^{m,p}(\Omega) \equiv \{u \in L^p(\Omega) : D^\alpha u \in L^p(\Omega) \text{ for } 0 \leq |\alpha| \leq m\}$ where $D^\alpha u$ is the weak partial derivative
- $W_0^{m,p}(\Omega) \equiv$ the closure of $C_0^\infty$ in the space $W^{m,p}(\Omega)$

It can be shown by Theorem 3.17 of [17] that for $1 \leq p < \infty$ we have

$$H^{m,p}(\Omega) = W^{m,p}(\Omega) \tag{2.45}$$

so for the rest of this section, we will use the space $H^{m,p}(\Omega)$.

### 2.4.2 Fractional Sobolev Spaces

Suppose we have a general, possibly nonsmooth, open set $\Omega \subset \mathbb{R}^n$. For any real $s > 0$ and for any $p \in [1, \infty)$, we aim to define the fractional Sobolev spaces $W^{s,p}(\Omega)$.

To construct this, let's first fix the fractional exponent $s \in (0, 1)$. Then for any $p \in [1, +\infty)$ we define $W^{s,p}(\Omega)$ as [18]:

$$W^{s,p}(\Omega) := \left\{ u \in L^p(\Omega) : \frac{|u(x) - u(y)|}{|x - y|^{\frac{n}{p}+s}} \in L^p(\Omega \times \Omega) \right\} \tag{2.46}$$

which can be interpreted as an intermediary Banach space between $L^p(\Omega)$ and $W^{1,p}(\Omega)$ endowed with the norm

$$\|u\|_{W^{s,p}(\Omega)} := \left( \int_\Omega |u|^p \, \mathrm{d}x + \int_\Omega \int_\Omega \frac{|u(x) - u(y)|^p}{|x - y|^{n+sp}} \, \mathrm{d}x\mathrm{d}y \right)^{\frac{1}{p}} \tag{2.47}$$

where we have used the *Gagliardo seminorm* of $u$:

$$[u]_{W^{s,p}(\Omega)} := \left( \int_\Omega \int_\Omega \frac{|u(x) - u(y)|^p}{|x - y|^{n+sp}} \, \mathrm{d}x\mathrm{d}y \right)^{\frac{1}{p}} \tag{2.48}$$

### 2.4.3 Methodology

# Chapter 3

# Black-Scholes Model

The Black-Scholes model [1] is the fundamental model for option pricing, which assumes that the underlying asset price follows a geometric Brownian motion. As with the market introduced in section (2.1.3), the market consists of two underlying assets:

- The risk-free asset, described by a deterministic function

$$\mathrm{d}B_t = rB_t\mathrm{d}t \tag{3.1}$$

  where for convenience, we set $B_0 = 1$ and $r$ is the constant risk-free interest rate with continuous compounding. This has the solution $B_t = e^{rt}$ for $t \geq 0$.

- The risky asset $S_t$, which has the price process of a geometric Brownian motion (GBM), which satisfies the stochastic differential equation

$$\mathrm{d}S_t = \mu S_t\mathrm{d}t + \sigma S_t\mathrm{d}W_t \qquad 0 \leq t \leq T \tag{3.2}$$

  for a given drift $\mu$ and volatility $\sigma$. We can solve this SDE to yield the unique solution

$$S_t = S_0 \exp\left(\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right) \qquad 0 \leq t \leq T \tag{3.3}$$

Under risk-neutral pricing, we can find an equivalent probability measure $\mathbb{Q}$ under which the discounted stock price is a martingale, and hence all assets earn the risk-free rate in expectation after discounting. Under the risk-neutral measure, the SDE for the risky asset becomes

$$\mathrm{d}S_t = rS_t\mathrm{d}t + \sigma S_t\mathrm{d}W_t^{\mathbb{Q}} \tag{3.4}$$

where the drift has been replaced by the risk-free rate $r$. The derivation can be found in Appendix A.1.

## 3.1 Closed-form solution for European options

Our first benchmark will be the closed-form solution for the European option, which would be used to evaluate the accuracy of the binomial tree later. The derivation of this formula is widely available and hence I will only include the main ideas. Using the risk-neutral pricing formula, the value of the European call option with strike $K$ and expiry $T$ is given by

$$v(0, S_0) = e^{-rT}\mathbb{E}^{\mathbb{Q}}\left[(S_T - K)^+ \mid S_0\right] \tag{3.5}$$

We use the solution of geometric brownian motion (3.3) to note that

$$S_T = S_0 \exp\left((r - \tfrac{1}{2}\sigma^2)T + \sigma W_T^{\mathbb{Q}}\right) \tag{3.6}$$

under the risk-neutral measure $\mathbb{Q}$, noting we have replaced the drift $\mu$ with the risk-free rate $r$. The Brownian motion is essentially the standard normal variable $Z \sim \mathcal{N}(0, 1)$ so we have

$$S_T = S_0 \exp\left((r - \tfrac{1}{2}\sigma^2)T + \sigma Z\right) \tag{3.7}$$

This can be plugged into the risk-neutral pricing formula (3.5): the exercise event $\{S_T > K\}$ is equivalent to $\{Z > -d_2\}$ where

$$d_2 = \frac{\log(S_0/K) + (r - \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \qquad d_1 = d_2 + \sigma\sqrt{T} \tag{3.8}$$

The expectation (3.5) can be split

$$\begin{aligned} v(0, S) &= e^{-rT}\mathbb{E}^{\mathbb{Q}}[(S_T - K)^+] && (3.9)\\ &= e^{-rT}\mathbb{E}^{\mathbb{Q}}[(S_T - K)\mathbf{1}_{\{S_T > K\}}] && (3.10)\\ &= e^{-rT}\mathbb{E}^{\mathbb{Q}}[S_T\mathbf{1}_{\{S_T > K\}}] - Ke^{-rT}\mathbb{Q}(S_T > K) && (3.11) \end{aligned}$$

The second probability is immediate as we see $\mathbb{Q}(S_T > K) = \mathbb{Q}(Z > d_2) = \Phi(d_2)$ by symmetry, where $\Phi$ is the CDF of the standard normal. We can complete the square in the Gaussian integrand for the first expectation to get the final formula [1]

$$v(0, S_0) = S_0\Phi(d_1) - Ke^{-rT}\Phi(d_2) \tag{3.12}$$

## 3.2 Binomial Tree Method

We can use the binomial discrete-time model to price both European and American options with the tree method. This model uses the simplification that the stock price can only move to one of two possible new prices in a time step. To avoid explosion of nodes, we will only consider recombinant trees so that the algorithm performs with quadratic time complexity. We will verify the approximated solution's validity against the closed-form formula for European options, and use its American variant as a benchmark for later methods.

To formalise the tree construction, let's fix a time horizon $T > 0$. For some depth $n \geq 1$, divide the interval $[0, T]$ into $n$ even steps each of length $\Delta t = T/n$. Our asset $S_t$ will follow a multiplicative binomial evolution:

$$S_{i+1} = \begin{cases} S_i u & \text{with probability } p \\ S_i d & \text{with probability } 1 - p \end{cases} \qquad S_0 > 0 \tag{3.13}$$

where $u$ and $d$ are the model parameters determining the two factors of the stock price change per time step.

As with (3.1), the financial market will also consist of a risk-free asset $B_t$, discretised so that it satisfies

$$B_{i+1} = B_i e^{r\Delta t}, \qquad B_0 = 1 \tag{3.14}$$

where $r$ is the continuously-compounded risk-free rate, typically considered interest. For an arbitrage free market, we should additionally ensure that

$$d < e^{r\Delta t} < u \tag{3.15}$$

for sufficiently small $\Delta t$, else we can make a riskless profit by shorting the stock and investing in the risk-free asset, or vice versa.

To employ risk-neutral pricing, we additionally require the discounted asset price to have zero drift (admitting a martingale measure). More precisely, let the discounted price be

$$\tilde{S}_i = e^{-ri\Delta t} S_i \tag{3.16}$$

We require some probability measure $\mathbb{Q}$, equivalent to the physical measure $\mathbb{P}$ (have the same non-zero support) such that

$$\mathbb{E}_{\mathbb{Q}}[\tilde{S}_{i+1} | \mathcal{F}_i] = \tilde{S}_i \tag{3.17}$$

To calculate the risk-neutral probability, we can apply this to our model:

$$e^{-r(i+1)\Delta t} \big[ p S_i u + (1-p) S_i d \big] = e^{-ri\Delta t} S_i \tag{3.18}$$

Collect the $p$ terms and cancel out the discount factor to get

$$p(S_i u - S_i d) + S_i d = e^{r\Delta t} S_i \tag{3.19}$$

Divide both sides by $S_i$ and rearrange to make $p$ the subject to obtain

$$p = \frac{e^{r\Delta t} - d}{u - d} \tag{3.20}$$

This is what we will use in our binomial model.

To price the option at time $0$, we will perform backward induction from expiry, since we know that the price is just the payoff $g : \mathbb{R}^+ \to \mathbb{R}$ as the option expires.

As the discounted price process is a martingale under the risk-free measure $\mathbb{Q}$, the arbitrage-free value $v(i\Delta t)$ of the derivative at time $i\Delta t$ is necessarily

$$v_i = \underbrace{e^{-r(T-i\Delta t)}}_{\text{discount}} \mathbb{E}_{\mathbb{Q}}[g(S_T) \,|\, \mathcal{F}_i] \tag{3.21}$$

We will utilise the Cox-Ross-Rubinstein (CRR) Model [3] for our implementation, which chooses the parameters

$$u = e^{\sigma\sqrt{\Delta t}} \qquad d = e^{-\sigma\sqrt{\Delta t}} \tag{3.22}$$

satisfying the arbitrage free condition, and normalising the local variance of returns to the volatility parameter $\sigma^2$. They also conveniently multiply to $1$.

To implement this, we need to first compute the price tree. This would be, for the price at step $i$ after $j$ up-moves (so clearly $i \geq j$)

$$S_{i,j} = S_0 u^j d^{i-j} \tag{3.23}$$

As the payoff is known at maturity, we can simply set

$$v_n(j) = g(S_{n,j}) \qquad 0 \leq j \leq n \tag{3.24}$$

and recursively set

$$v_i(j) = e^{-r\Delta t} \big( p^* v_{i+1}(j+1) + (1-p^*) v_{i+1}(j) \big) \tag{3.25}$$

for $0 \leq j \leq i$. This would be it for European options, but since early exercise is allowed, we will compare this value with the payoff of the price at step $i$ with $j$ up-moves, which is just $f(S_{i,j})$. Thus the value is given by

$$v_i(j) = \max\left( e^{-r\Delta t} \big( p^* v_{i+1}(j+1) + (1-p^*) v_{i+1}(j) \big),\ g(S_{i,j}) \right) \tag{3.26}$$

The root value $v_0(0)$ would be the arbitrage-free option price.

## 3.3 Physics-Informed Neural Networks

In order to use the framework of PINNs as detailed in section 2.3, we need to first derive the PDE satisfied by the option price, as well as the boundary conditions. We will first work with the one-dimensional case, then discuss the extension to the multi-dimensional case and how we could build a benchmark to against which we evaluate our results.

### 3.3.1 PINN formulation for one-dimensional case

In order to set up our PINN, we need to first derive the PDE satisfied by the option price. Recall the value function $v : [0,T] \times \mathbb{R}^+ \to \mathbb{R}$ of an American put with payoff $g : \mathbb{R}^+ \to \mathbb{R}$ at time $t$ with asset price $S_t = S$ is given by

$$v(t,S) = \sup_{\tau \in [t,T]} \mathbb{E}_t\big[e^{-r(\tau - t)} g(S_\tau) \mid S_t = S\big] \tag{3.27}$$

where $\tau$ is any stopping time in $[t,T]$ and $\mathbb{E}_t$ is conditional expectation at time $t$, given $S_t = S$. For an American option, we always have two options:

- Immediately exercise the option and get payoff $g(S)$. Since this is not necessarily optimal, the payoff must bound the value option from below:

$$v(t,S) \geq g(S) \tag{3.28}$$

- Do nothing and let the option continue to time $t+h$ with $h > 0$ a very small number. By using the supremum definition of $v(t,S)$ (3.27) and swapping the expectation and supremum [19] we arrive at

$$e^{rh} v(t,S) \geq \mathbb{E}_t\big[v(t+h, S_{t+h})\big] \tag{3.29}$$

  and the inequality turns into an equality at the limit $\delta \to 0$.

We will additionally assume that the value function $v \in C^{1,2}$ (continuously differentiable with respect to $t$ and twice continuously differentiable with respect to $S$) so that we can apply Itô's formula. Let's model the asset price $S$ as a geometric Brownian motion satisfying under the risk-neutral measure

$$\mathrm{d}S_t = rS_t \mathrm{d}t + \sigma S_t \mathrm{d}W_t \tag{3.30}$$

Then by Ito's lemma and taking expectations of both sides we have

$$\mathbb{E}\big[v(t+\delta, S_{t+h})\big] = v(t,S) + \mathbb{E}_t\left[\int_t^{t+h} \left(\frac{\partial v}{\partial t}(u,S_u) + \frac{\partial v}{\partial S}(u,S_u) r S_u + \frac{1}{2}\frac{\partial^2 v}{\partial S^2}(u,S_u)\sigma^2 S_u^2\right) \mathrm{d}u\right] \tag{3.31}$$

Now let's substitute (3.29) into the LHS, which gives us

$$e^{rh} v(t,S) \geq v(t,S) + \mathbb{E}_t\left[\int_t^{t+h} \left(\frac{\partial v}{\partial t}(u,S_u) + \frac{\partial v}{\partial S}(u,S_u) r S_u + \frac{1}{2}\frac{\partial^2 v}{\partial S^2}(u,S_u)\sigma^2 S_u^2\right) \mathrm{d}u\right] \tag{3.32}$$

We can use the Taylor approximation $e^{rh} = 1 + rh + o(h)$, which permits us to cancel the $v(t, S)$ on both sides. Dividing both sides by $h$ in the same step, we get

$$rv(t,S) + \frac{o(h)}{h} \geq \mathbb{E}_t\left[\frac{1}{h}\int_t^{t+h}\left(\frac{\partial v}{\partial t}(u,S_u) + \frac{\partial v}{\partial S}(u,S_u)rS_u + \frac{1}{2}\frac{\partial^2 v}{\partial S^2}(u,S_u)\sigma^2 S_u^2\right)\mathrm{d}u\right] \tag{3.33}$$

Sending $h \to 0$ makes the higher order term $o(h)/h$ vanish, and the expectation of the integral to collapse to the integrand. Thus we obtain

$$rv(t,S) \geq \frac{\partial v}{\partial t}(t,S) + \frac{\partial v}{\partial S}(t,S)rS + \frac{1}{2}\frac{\partial^2 v}{\partial S^2}(t,S)\sigma^2 S^2 \tag{3.34}$$

or for simplicity, we define the linear parabolic operator $\mathcal{L}$ [8]:

$$\mathcal{L}[v] = rv(t,S) - \frac{\partial v}{\partial S}(t,S)rS - \frac{1}{2}\frac{\partial^2 v}{\partial S^2}(t,S)\sigma^2 S^2 \tag{3.35}$$

Thus we can concisely express this inequality as

$$\min\left(-\frac{\partial v}{\partial t}(t,S) + \mathcal{L}[v], v(t,S) - g(S)\right) \geq 0 \tag{3.36}$$

However, note that at time $t$, we must either exercise the option or do nothing and continue. One of these two options must be optimal, implying that one of the arguments must be $0$. Thus we actually have an equality:

$$\min\left(-\frac{\partial v}{\partial t}(t,S) + \mathcal{L}[v], v(t,S) - g(S)\right) = 0 \tag{3.37}$$

which is the variational equation [20] for the American option. Note this is exactly the formulation of the PINN problem (2.36).

We are now ready to derive the boundary conditions. Our option has a terminal payoff at time $T$ given by $g(S)$, so we have the terminal condition

$$v(T,S) = g(S) \tag{3.38}$$

which satisfies the form of the terminal condition for PINNs (2.37). For the spatial boundary conditions, we have two boundaries to consider: $S \to 0$ and $S \to \infty$. In practice, we will have to truncate the domain to $[S_{\text{min}}, S_{\text{max}}]$ for some sufficiently small $S_{\text{min}}$ and sufficiently large $S_{\text{max}}$. At the lower boundary, we have the boundary condition

$$v(t,S_{\text{min}}) = g(S_{\text{min}}) \tag{3.39}$$

because it is always optimal to exercise the option when the asset is at its lowest price. At the upper boundary, we assume the option is too far out of the money to ever finish in the money, and hence the option price is approximately $0$. Thus we have the boundary condition

$$v(t,S_{\text{max}}) = 0 \tag{3.40}$$

These three conditions together would form the boundary setup (2.38) for our PINN.

### 3.3.2 Reduction of multi-asset to one-asset case

Before we apply the PINN framework to the multi-dimensional case, we will first introduce a benchmark for the multi-dimensional case, from which we will build the boundary conditions for the PINN, as they change depending on the payoff function. The benchmark will need to be able to be reduced to a one-dimensional problem, so that we can solve it using the binomial tree method as a reference for our PINN results. This will, in turn, give us confidence that our PINN implementation is correct, and that the results we get for more complex payoffs are reasonable.

In the multi-dimensional case, we have $k$ underlying assets with price processes following a correlated geometric Brownian motion. Mathematically, let each asset $S^i$ satisfy under the risk-free measure $\mathbb{Q}$, for $i = 1, \dots, k$,

$$\mathrm{d}S_t^i = rS_t^i\mathrm{d}t + \sigma_i S_t^i \mathrm{d}W_t^i \tag{3.41}$$

with $\langle W^i, W^j\rangle_t := \mathbb{E}\big[\mathrm{d}W_t^i \mathrm{d}W_t^j\big] = \rho_{ij}\mathrm{d}t$ for all $i, j$ pairs. As the geometric Brownian motion is multiplicative, for our benchmark, we will consider the put on the product of the assets. More precisely, we consider the put option with payoff

$$g(S^1, \dots, S^k) = \left(K - \prod_{i=1}^{k} S^i\right)^+ \tag{3.42}$$

To represent the dynamics of the product of the assets, we can apply Ito's formula to the log of the price process. The log representation is

$$\ln S_t^i = \ln S_0^t + \int_0^t \left(r - \frac{1}{2}\sigma_i^2\right)\,\mathrm{d}s + \int_0^t \sigma_i\,\mathrm{d}W_s^i \tag{3.43}$$

Define our new asset $X$ to be

$$X_t := \prod_{i=1}^{k} S_t^i \tag{3.44}$$

and hence we have

$$\ln X_t = \sum_{i=1}^{k} \ln S_t^i = \ln X_0 + \int_0^t \left(kr - \frac{1}{2}\sum_{i=1}^{k}\sigma_i^2\right)\,\mathrm{d}s + \sum_{i=1}^{k}\int_0^t \sigma_i\,\mathrm{d}W_s^i \tag{3.45}$$

By taking the differential we have

$$\mathrm{d}\ln X_t = \left(kr - \frac{1}{2}\sum_{i=1}^{k}\sigma_i^2\right)\,\mathrm{d}t + \sum_{i=1}^{k}\sigma_i\,\mathrm{d}W_s^i \tag{3.46}$$

The quadratic variation of the martingale term is

$$\left\langle \sum_{i=1}^{k}\sigma_i W^i, \sum_{j=1}^{k}\sigma_j W^j \right\rangle_t = \sum_{i=1}^{k}\sum_{j=1}^{k}\sigma_i\sigma_j\rho_{ij}t =: \sigma_X^2 t \tag{3.47}$$

i.e. $\sigma_X^2 = \sum_{i=1}^{k}\sum_{j=1}^{k}\sigma_i\sigma_j\rho_{ij}$. So we can express the second term in terms of a new Brownian motion $\widetilde{W}$:

$$\sum_{i=1}^{k}\sigma_i\,\mathrm{d}W_s^i = \mathrm{d}\widetilde{W}_t \tag{3.48}$$

Going back, we have

$$\mathrm{d}\ln X_t = \left(kr - \frac{1}{2}\sum_{i=1}^{k}\sigma_i^2\right)\mathrm{d}t + \sigma_X \mathrm{d}\widetilde{W}_t \tag{3.49}$$

Exponentiating using Ito's formula gets us

$$\mathrm{d}X_t = X_t\left[\left(kr - \frac{1}{2}\sum_{i=1}^{k}\sigma_i^2\right)\mathrm{d}t + \sigma_X \mathrm{d}\widetilde{W}_t\right] + \frac{1}{2}X_t\sigma_X^2\mathrm{d}t \tag{3.50}$$

which we can rearrange to get

$$\mathrm{d}X_t = \left(kr - \frac{1}{2}\sum_{i=1}^{k}\sigma_i^2 + \frac{1}{2}\sigma_X^2\right)X_t\mathrm{d}t + \sigma_X X_t\,\mathrm{d}\widetilde{W}_t \tag{3.51}$$

which is a geometric Brownian motion with volatility $\sigma_X$ and drift $r_X$ where

$$r_X = kr - \frac{1}{2}\sum_{i=1}^{k}\sigma_i^2 + \frac{1}{2}\sigma_X^2 \tag{3.52}$$

$$= kr + \sum_{i<j}\sigma_i\sigma_j\rho_{ij} \tag{3.53}$$

by matching coefficients. We can verify the equivalence by simulating many price paths for the assets and comparing their product against the simulations of price paths for the equivalent asset. Using the initial values

$$r = 0.05 \qquad S_0 = (10, 5, 3)^\top \qquad \sigma = (0.2, 0.3, 0.1)^\top \qquad \mathrm{R} = \begin{pmatrix} 1 & 0.6 & -0.2 \\ 0.6 & 1 & 0.4 \\ -0.2 & 0.4 & 1 \end{pmatrix} \tag{3.54}$$

where $\mathrm{R}$ is the correlation matrix, we can plot the histogram of final prices in Figure [3.1].

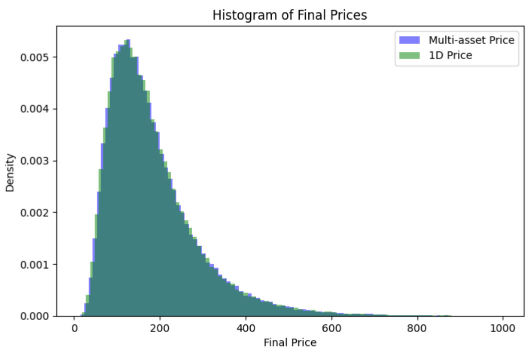


**Figure 3.1** Histogram of final prices for product of assets and equivalent asset

They look very similar, and we can further statistically test if they come from the same distribution. Using Kolmogorov-Smirnov's 2-sample test on the null hypothesis that they come from the same distribution, we obtain a p-value of $0.8491$, which does not provide sufficient evidence to reject the null hypothesis, which gives us stronger evidence that our derivation is correct.

Therefore, we will be able to price the put option on the product of multiple assets and compare that to the price of the put option on this single constructed asset as a benchmark.

### 3.3.3 PINN formulation for multi-dimensional case

With the benchmark in place, we now need to derive the variational inequality for the multi-dimensional case where we have $k$ assets. This is a straightforward extension of the one-dimensional case, where the value function $v(t, S^1, \ldots, S^k)$ now takes $k+1$ inputs. For notation, we will express the multiple assets as a vector $\mathbf{S} = (S^1, \ldots, S^k)^\top$, so the value function is $v(t, \mathbf{S})$. Similarly, the payoff function will be written as $g(S^1, \ldots, S^k) = g(\mathbf{S})$.

The value function stays

$$v(t, \mathbf{S}) = \sup_{\tau \in [t,T]} \mathbb{E}_t\big[e^{-r(\tau - t)} g(\mathbf{S}_\tau) \mid \mathbf{S}_t = \mathbf{S}\big] \tag{3.55}$$

Should we choose not to exercise the option at time $t$, we can apply the same argument as in the one-dimensional case: considering a small $h > 0$, we can apply Itô's lemma on multiple dimensions to yield

$$\begin{aligned} v(t+h, \mathbf{S}_{t+h}) = v(t, \mathbf{S}) + \int_t^{t+h} \bigg( &\frac{\partial v}{\partial t}(u, \mathbf{S}_u) + \sum_{i=1}^k \frac{\partial v}{\partial S^i}(u, \mathbf{S}_u) r S_u^i \\ &+ \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k \frac{\partial^2 v}{\partial S^i \partial S^j}(u, \mathbf{S}_u) \sigma_i \sigma_j \rho_{ij} S_u^i S_u^j \bigg) \, \mathrm{d}u \end{aligned} \tag{3.56}$$

The steps from here are identical to the one-dimensional case, where we can take expectations of both sides and use the second order Taylor expansion of the exponential to get the variational inequality

$$rv(t, \mathbf{S}) \geq \frac{\partial v}{\partial t}(t, \mathbf{S}) + \sum_{i=1}^k \frac{\partial v}{\partial S^i}(t, \mathbf{S}) r S^i + \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k \frac{\partial^2 v}{\partial S^i \partial S^j}(t, \mathbf{S}) \sigma_i \sigma_j \rho_{ij} S^i S^j \tag{3.57}$$

where to simplify into the PINN framework as in (2.36), we can define the operator $\mathcal{L}$ as

$$\mathcal{L}[v] = rv(t, \mathbf{S}) - \sum_{i=1}^k \frac{\partial v}{\partial S^i}(t, \mathbf{S}) r S^i - \frac{1}{2} \sum_{i=1}^k \sum_{j=1}^k \frac{\partial^2 v}{\partial S^i \partial S^j}(t, \mathbf{S}) \sigma_i \sigma_j \rho_{ij} S^i S^j \tag{3.58}$$

and the problem reduces to the same variational inequality as in the one-dimensional case (3.37), with a different operator $\mathcal{L}$.

To finish the PINN formulation, we need to derive the boundary conditions. The terminal condition is straightforward, as it is given by the payoff function:

$$v(T, \mathbf{S}) = g(\mathbf{S}) = \left( K - \prod_{i=1}^k S^i \right)^+ \tag{3.59}$$

For the spatial boundary conditions, we will again truncate the domain to

$$\Omega = \Big\{ \mathbf{S} = (S^i, \ldots, S^k) \in \mathbb{R}^k : S^i \in [S^i_{\text{min}}, S^i_{\text{max}}] \text{ for } i = 1, \ldots, k \Big\} \tag{3.60}$$

for some sufficiently small and large respective minimum and maximum values for each stock. The lower boundary will be sampled from the set

$$\partial_{\text{lower}} \Omega = \big\{ \mathbf{S} \in \Omega : S^i = S^i_{\text{min}} \text{ for some } i \big\} \tag{3.61}$$

where we have the boundary condition

$$v(t, \mathbf{S}) = g(\mathbf{S}), \qquad \mathbf{S} \in \partial_{\text{lower}}\Omega \tag{3.62}$$

and the upper boundary will be sampled from the set

$$\partial_{\text{upper}}\Omega = \left\{\mathbf{S} \in \Omega : S^i = S^i_{\text{max}} \text{ for some } i\right\} \tag{3.63}$$

with the boundary condition

$$v(t, \mathbf{S}) = 0, \qquad \mathbf{S} \in \partial_{\text{upper}}\Omega \tag{3.64}$$

analogous to the one-dimensional case. Of course, this framework could be easily adapted to other payoff functions, where the boundary conditions would change accordingly (but not the variational inequality). But for our benchmark, this is the setup we will use.

## 3.4 Sobolev Deep Neural Networks

### 3.4.1 Model

Suppose we would like to price an option on $k$ assets $\boldsymbol{S} = (S^1, \dots, S^k)$. As asset prices are theoretically unbounded, we truncate it in our model. Thus the domain of the target function can be chosen to be

$$\Omega = (S^1_{\text{min}}, S^1_{\text{max}}) \times \cdots \times (S^k_{\text{min}}, S^k_{\text{max}}) \tag{3.65}$$

In our study we will make the simplification that $S^i_{\text{min}} = S^j_{\text{min}}$ and $S^i_{\text{max}} = S^j_{\text{max}}$ for all $i \neq j$, which we can call $a$ and $b$ respectively. Our spaces would consequently have the definitions

$$\Omega = (a, b)^n \qquad \Omega_T = (0, T) \times \Omega \qquad \Sigma = (0, T) \times \partial\Omega \tag{3.66}$$

where $\partial\Omega =: \Gamma$ is the boundary of $\Omega$, a hyper-rectangle. We will use the representation

$$\Gamma = \bigcup_{i=1}^{n} \big(\{x \in [a, b]^n : x_i = a\} \cup \{x \in [a, b]^n : x_i = b\}\big) \tag{3.67}$$

which is not the simplest, but it is useful as it is decomposed into faces, which is useful when we would like to use the norm derivative later. Note that in the one-dimensional case, this reduces simply to the points $\{a, b\}$.

### 3.4.2 Loss function

Our loss function would be a weighted sum of four loss components $\sum_{i=1}^{4} \lambda_i J_i$, each of which ensures regulation of a specific aspect of our function.

1. The variational loss, which is as before, controlling the accuracy of our function in the interior of our domain:

$$J_1 = \| \min\{\mathcal{G}[f], f - g\}\|^2_{L^2(\Omega_T), \mu_1} \tag{3.68}$$

2. The energy loss: before we were only controlling the L2 norm of the payoff, but we require the option value to equal the payoff as a function and not just pointwise. Thus we need to incorporate the gradient into the loss so that the spatial slope of the payoff is respected.

$$J_2 = \|f(0,\cdot) - g(\cdot)\|^2_{H^1(\Omega),\mu_2} \tag{3.69}$$

where we have the norm defined as

$$\|d\|^2_{H^1(\Omega),\mu_2} = \|d\|^2_{L^2(\Omega)} + \|\nabla d\|^2_{L^2(\Omega)} \tag{3.70}$$

and the derivative

$$\nabla d = \left(\partial_{x_1} d, \ldots, \partial_{x_k} d\right) \tag{3.71}$$

3. The Dirichlet trace loss:

$$J_3 = \|f(t,x) - g(x)\|^2_{H^{\frac{3}{4},\frac{3}{2}}(\Sigma),\mu_3} \tag{3.72}$$

where

$$\|d\|^2_{H^{\frac{3}{4},\frac{3}{2}}(\Sigma)} = \|d\|^2_{H^{0,1}(\Sigma)} + \int_0^T \int_0^T \frac{\|d(t_1,\cdot) - d(t_2,\cdot)\|^2_{L^2(\Gamma)}}{|t_1 - t_2|^{2.5}} \,\mathrm{d}t_1 \,\mathrm{d}t_2 \tag{3.73}$$

$$+ \int_0^T \int_\Gamma \int_\Gamma \frac{|d(t,x) - d(t,y)|^2}{\|x - y\|^2} \,\mathrm{d}\sigma(x)\mathrm{d}\sigma(y)\mathrm{d}t \tag{3.74}$$

The time exponent $2.5$ is equal to $2p+1$ where $p$ is the decimal part of the time fractional exponent, and the space exponent $2$ is equal to $2\theta + k - 1$ where $\theta$ is the decimal part of the space fractional exponent, and $k$ the dimension. Note that here $\mathrm{d}\sigma(x)$ and $\mathrm{d}\sigma(y)$ indicate the surface measures on $\Gamma$, but in practice we can treat these as $\mathrm{d}x$ and $\mathrm{d}y$.

4. The Neumann trace loss:

$$J_4 = \|\nabla(f-g)\cdot\nu\|^2_{H^{\frac{1}{4},\frac{1}{2}}(\Sigma),\mu_4} \tag{3.75}$$

where the norm derivative is $\nabla(f-g)\cdot\nu = \partial_{x_{i^\star}}(f-g)$ given that $x_{i^\star} \in \{a,b\}$ and $x_i \in (a,b)$ for $i \neq i^\star$. Rigorously, we require the sign but in practice this is not necessary, as we will square in the end. The fractional norm is thus written

$$\|d\|^2_{H^{\frac{1}{4},\frac{1}{2}}(\Sigma)} = \|d\|^2_{L^2(\Sigma)} + \int_0^T \int_0^T \frac{\|d(t_1,\cdot) - d(t_2,\cdot)\|^2_{L^2(\Gamma)}}{|t_1 - t_2|^{1.5}} \,\mathrm{d}t_1\mathrm{d}t_2 \tag{3.76}$$

$$+ \int_0^T \int_\Gamma \int_\Gamma \frac{|d(t,x) - d(t,y)|^2}{\|x - y\|^2} \,\mathrm{d}\sigma(x)\mathrm{d}(y)\mathrm{d}t \tag{3.77}$$

because $H^{0,0} = L^2$.

The third and fourth losses are those that enforce $f$ to solve the option pricing variational inequality with regularity in the time and space dimensions beyond pointwise [21]. In practice, we will compute a Monte Carlo estimate of these integrals by sampling pairs $(t_1, t_2)$ from the domain, evaluating the integrand and taking the mean across all pairs.

## 3.5 Numerical Results

### 3.5.1 Tree benchmark against closed-form solution

First we will evaluate the performance of the benchmark binomial tree method for the one-dimensional case against the closed-form solution, which we would take as the ground truth.

To do this, we will compute the tree option price without taking the maximum of the payoff and continuation value at each step so that we can compare the tree price against the Black-Scholes price.

First, we will evaluate the tree's accuracy on the most sensitive point, which is at time $t = 0$ and when the initial asset price $S$ is equal to the strike price $K$ (at the money). Figure 3.2 shows the convergence of the tree price to the closed-form Black-Scholes price for European put options as we increase the depth of the tree. We can see that around $1000$ steps yields a sufficiently close price to the closed-form solution, and further increasing the depth does not yield significant improvements, so this is the depth we will use for our tree benchmark.

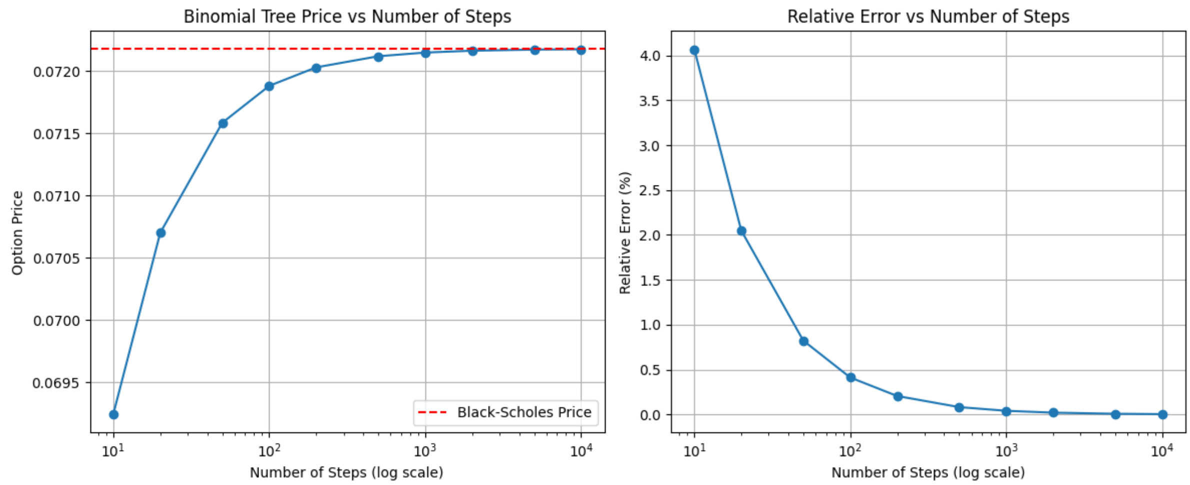


**Figure 3.2** Tree convergence at $t = 0, S = K$ over different depths

Evaluating the tree price against the closed-form solution for different initial asset prices $S$ at the initial time $t = 0$ in Figure 3.3, we can see that the tree price is very close to the ground truth, with less than $0.05\%$ of relative error until we are far out of the money where the option price is very close to $0$ and unsurprisingly yields greater errors, where machine precision becomes a limiting factor.

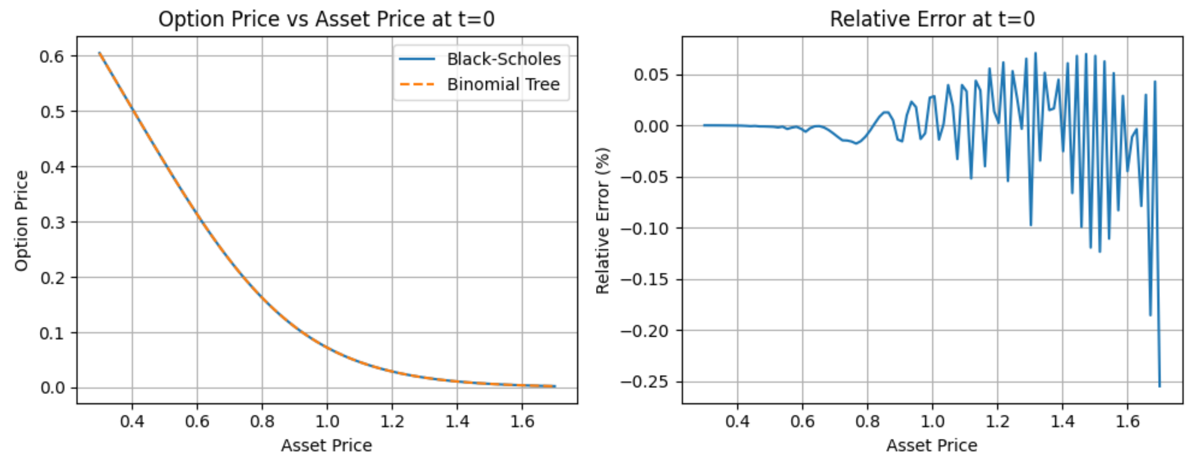


**Figure 3.3** Comparison of tree and closed-form prices for European put option with varying initial asset price

We can see that the tree price also closely tracks the term structure of the option price, as shown in Figure 3.4, where the error is again less than $0.05\%$, decreasing as we get closer to maturity, which is expected as the tree has not done as many backward-induction steps to get to the initial time, where errors can accumulate.

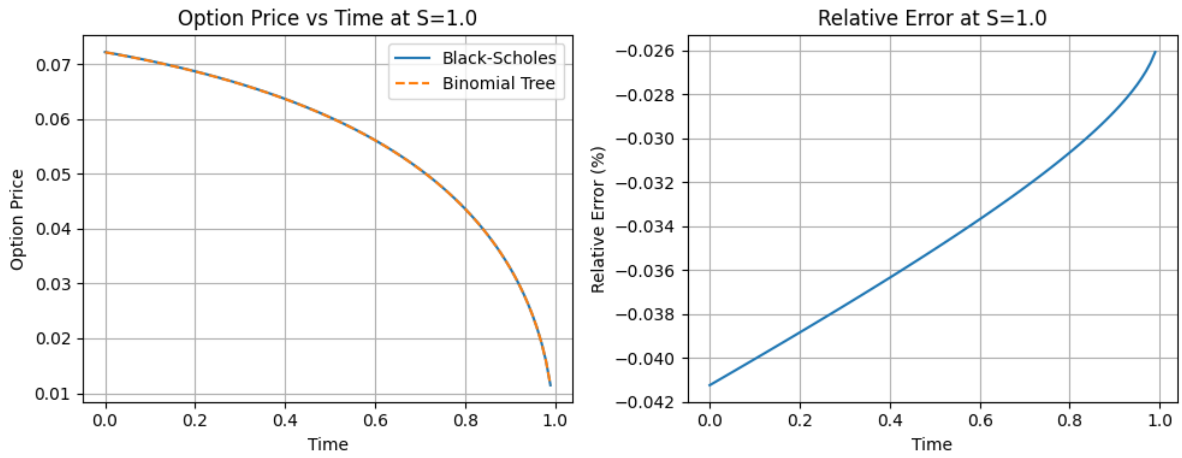


**Figure 3.4** Comparison of tree and closed-form prices for European put option with varying time

Overall, we can conclude that the tree benchmark yields very strong results against the closed-form solution, giving us good confidence that it will provide accurate results for American options too, where we do not have a closed-form solution to compare against. This will serve as our benchmark for evaluating the performance of our PINN implementation for American options.

### 3.5.2 PINN results against tree benchmark

We will now train our PINN on the one-dimensional case and compare the results against the tree benchmark on the American put on the asset with parameters $K = 1, r = 0.1, \sigma = 0.3, T = 1$ across a range of possible initial asset prices $S_0 \in [0, 3]$. This will be done through a feedforward neural network with 3 hidden layers each of width 32, with learning rate 0.001, batch size 10000 with 30000 maximum epochs with early stopping patience of 500 epochs. We train 25 models with different random initialisation and compute the ensemble mean for our final price estimate.

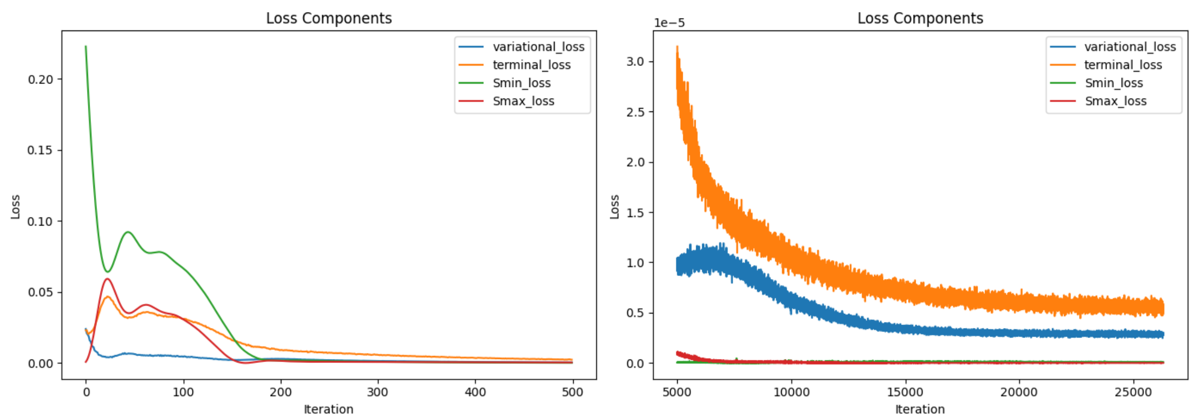


**Figure 3.5** Loss convergence for the one-dimensional PINN for one run

Figure 3.5 shows the convergence of the loss function for one run, and we can see that the loss converges well, with each of them having similar magnitudes, converging in the order of $10^{-5}$.

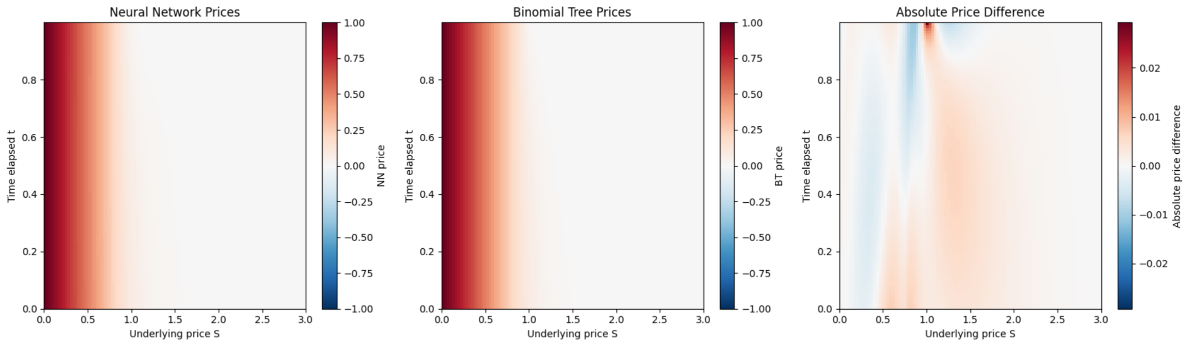


**Figure 3.6** PINN vs tree price comparison heatmap across domain

Figure 3.6 shows the heatmap of the PINN price against the tree price across the domain of time and initial asset price. We can see that the PINN price is very close to the tree price across the domain, where the time value is correctly captured at the most sensitive point of $S = K$ and $t = 0$ as seen in Figure 3.7. This figure additionally shows the model capturing the term structure of the option price, where the PINN price closely tracks the tree price across time for $S = K$.

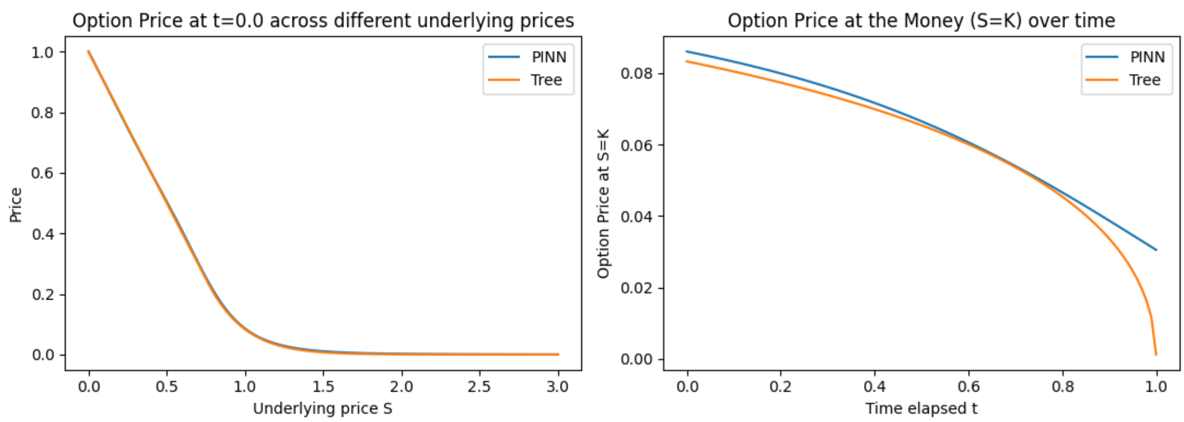


**Figure 3.7** Price comparison for varying underlying price, and for term structure at $S = K$

We can also zoom in on the most sensitive point of $(t, S) = (0, K)$ and evaluate the error. Table 3.1 shows the mean PINN price across 30 runs with its 95% confidence interval, and we can see that the relative error is around $3\%$, which is good considering we were sampling across the whole domain and not just at this point. However, we can see that the benchmark tree price lies around 4 standard deviations away from the PINN mean price, suggesting that there is some bias in the PINN approximation.

| Tree price | PINN price | Relative error |
|---|---|---|
| 0.0833 | 0.0860 $\pm$ 0.0006 | 3.33% |

**Table 3.1** PINN price at the money and initial time against tree benchmark

We can investigate this apparent bias by tracking the price $f(t = 0, S = K)$ across different training epochs, as shown in Figure 3.8. We can see that the price spikes above the tree price and then settles around the marked benchmark tree price but interestingly approaches it from below, confirming our suspicion that the PINN price is biased against the tree price. That said, the tree price is not guaranteed to be the ground truth.

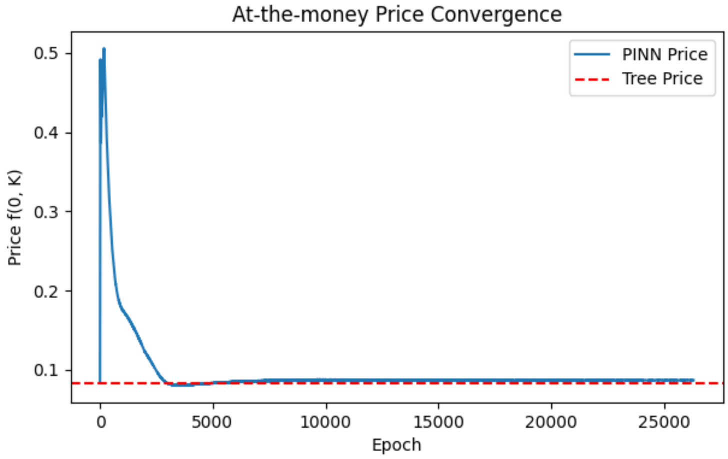


**Figure 3.8** $f(0, K)$ across training epochs for one run

### 3.5.3 Multi-dimensional PINN

We move onto training a PINN on multiple assets, for simplicity and visualisation purposes, we will train a two-dimensional PINN on the put option on the product of two assets, with parameters $K = 1, r = 0.1, T = 1$ with stock volatilities $\sigma = (0.3, 0.2)^\top$ and correlation between assets $\rho = 0.5$. Both of these assets will be truncated to $[0, 3]$ and we will sample uniformly from the whole domain for training. The model architecture will be similar to the one-dimensional case, with a bigger feedforward neural network of 4 hidden layers each of width 64, learning rate 0.001 and batch size 10000. This will be trained for a maximum of 40000 epochs with early stopping patience of 500 epochs across 10 different seeds. Figure 3.9 shows the heatmap of the PINN price against the tree price, where we take the $S_1$-$S_2$ cross section and fix time at $t = 0$. We can see that the PINN price is close to the tree price across the domain, with some stronger bias at the in-the-money regions where the price is higher and hence is not as significant when we take relative error into consideration.

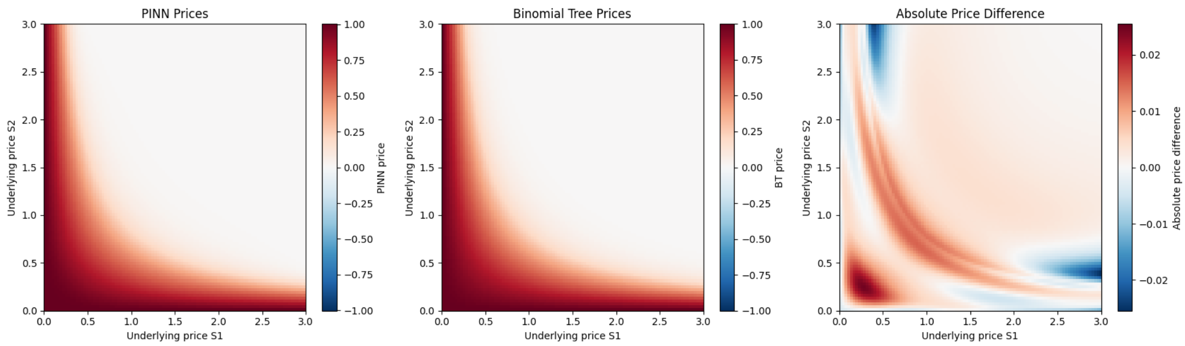


**Figure 3.9** PINN vs tree price comparison heatmap across $S_1$-$S_2$ cross section at $t = 0$

We can zoom in on the $S_1 \times S_2 = K$ cross section that we can see clearly in the heatmap, and plot the price along this cross section across time, shown in the left figure of Figure 3.7. We can see that the PINN fit is satisfactory, with inconsistencies at the tails due to our boundary conditions, which are set at $S = 0$ and $S = 3$ for both assets, which are not the true boundaries of the problem and a model simplication we made.

Overall, by Table 3.2, we can see that the relative error at the most sensitive point at the triple $(0, \sqrt{K}, \sqrt{K})$ is around 7% which is considerably higher than the one-dimensional case, which

| Tree price | PINN price | Relative error |
|---|---|---|
| 0.0993 | $0.1060 \pm 0.0048$ | 6.75% |

**Table 3.2** Prices $f(0, \sqrt{K}, \sqrt{K})$ for the two-dimensional PINN against tree benchmark

is expected as we are now sampling across a much larger domain, and we are restrained by model capacity and training time due to personal hardware constraints, which is a common issue for multi-dimensional PINNs.

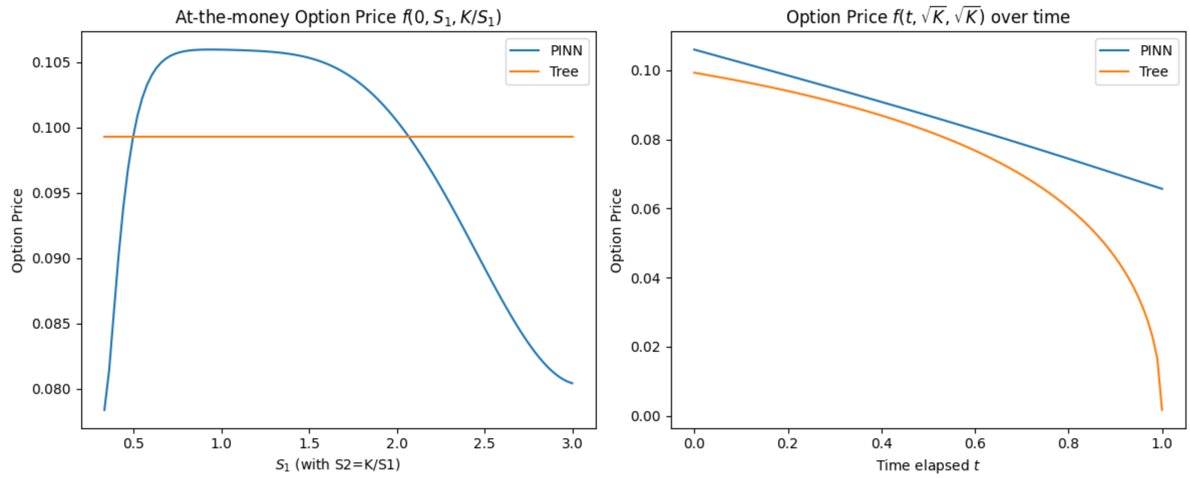


**Figure 3.10** PINN vs tree price comparison for at the money cross section and term structure

We can also take the cross section at $S_1 = S_2$ to evaluate the term structure of the option price, as shown in the right figure of Figure 3.10. We can see that the PINN has managed to capture the theta decay of the option price, but the consistent bias of the price is evident here and the model struggles to capture the more rapid theta decay as we approach maturity. This may be due to the fact gamma approaches a dirac delta function at the money as we approach maturity, whose sharp feature is difficult for the model to capture when our samples are sparser in the expanded domain.

### 3.5.4 Free boundary

#### One-dimensional case

For the one-dimensional free boundary, we will first evaluate the free boundary from the tree benchmark. Here, the continuation values are extracted through a small modification to the tree algorithm, where we compute the price of the Bermudan option with the same parameters with possibility of exercise at every time step except for at $t = 0$, effectively serving as a continuation value of the American option at $t = 0$. This continuation value is compared with the intrinsic value (the payoff) like we introduced in equation (2.9) in the mathematical background. The left figure in Figure 3.11 shows the continuation probabilities across different $(t, S)$ pairs, and here we use hard classification as binomial tree pricing is deterministic.

We can see that the free boundary is correctly captured at glance, where we exercise the option when the asset price is sufficiently low, whose threshold increases as we get closer to maturity, consistent with the intuition that the option is losing time value.

We now evaluate the free boundary from the PINN. The central figure in Figure 3.11 shows the continuation probabilities across different $(t, S)$ pairs using the sigmoid distribution as described in equation (2.12). We can see that the shape of the free boundary is similar to the tree benchmark, and with soft classification the lowest probability is around $0.4$, which is not unintuitive as in the exercise region, the continuation value is theoretically equal to the payoff, and hence we should be indifferent between exercising and continuing.

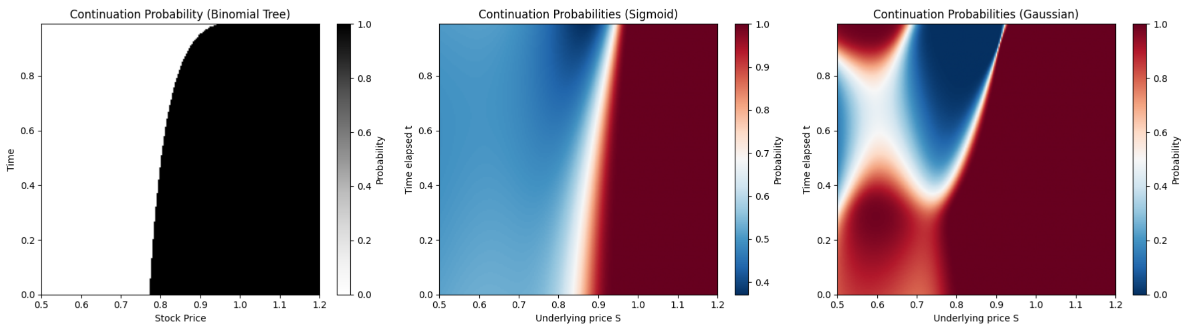


**Figure 3.11** Estimated continuation probabilities from PINN for the American put option

To use the Gaussian method, we will first evaluate its validity by plotting the distribution of the standardised price estimates across all seeds in Figure 3.12, noting that as the variance is unknown and we use the sample variance, the standardised price estimates would follow a t-distribution with $n-1$ degrees of freedom where $n$ is the number of seeds.

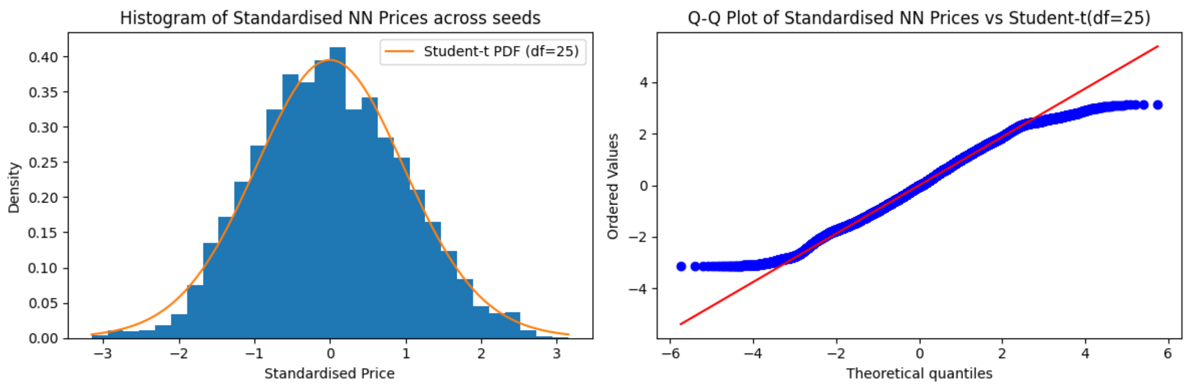


**Figure 3.12** Distribution of standardised neural network price estimates across 25 runs

By looking at the histogram and the Q-Q plot, we can see that the distribution of the standardised price estimates is reasonably close to a t-distribution, with the exception of the tails. Nevertheless, we will proceed with evaluating the free boundary using the Gaussian method, and the right figure in Figure 3.11 shows the continuation probabilities across different $(t, S)$ pairs.

We can see that the shape roughly corresponds to the free boundaries we have seen before, but this is a much noisier estimate, which puts into question the assumptions we have made. We are essentially assuming that the price estimates of the PINN are unbiased and normally distributed around the true price, which is quite strong and not true in practice, as we have already observed in Figure 3.6 where we have zones where the PINN price is systematically above or below the tree price, suggesting bias.

#### Two-dimensional case

# Chapter 4

# The Heston Model

The classic Black Scholes model makes the strong assumption that the stock returns are normally distributed with known mean and variance. However, the Black Scholes formula does not depend on the mean, so allowing the mean to vary does not help us to generalise the model. But if we allow volatility to vary according to a stochastic process, we obtain a more generalised model. Heston [2] proposes a model that is not based on the Black Scholes formula that provides a closed-form solution for European call options. In this model, the dynamics under the pricing measure of the asset price $S$ and the variance (squared volatility) process $V$ are governed by the stochastic differential equation system under the risk-neutral measure $\mathbb{Q}$:

$$\mathrm{d}S_t = rS_t\mathrm{d}t + \sqrt{V_t}\mathrm{d}W_t^1 \tag{4.1}$$

$$\mathrm{d}V_t = \kappa(\theta - V_t)\mathrm{d}t + \sigma\sqrt{V_t}\mathrm{d}W_t^2 \tag{4.2}$$

where we have initial conditions $S_0 = s > 0$ and $V_0 = v \geq 0$, and the two Brownian motions are correlated with

$$\mathrm{d}\langle W^1, W^2\rangle_t = \rho\mathrm{d}t \qquad \rho \in (-1, 1) \tag{4.3}$$

The dynamics of the variance $V$ follow a CIR process with mean reversion rate $\kappa > 0$, long run state $\theta > 0$ and volatility of the volatility $\sigma > 0$. We can enforce the Feller condition $2\kappa\theta \geq \sigma^2$ so that the volatility process stays positive.

## 4.1 Closed-form Solution

There exists a closed-form solution derived by Heston [2] to the price of the European call option. This will serve as a benchmark for our other techniques, which would give us confidence in the accuracy of our other techniques for European options, and by extension, for American options too.

The closed-form solution is derived by using an Ansatz inspired by the Black-Scholes formula, and determining the elements of our guess by substituting it into the PDE we know the solution must satisfy along with the boundary conditions. We will include a sketch of the derivation here which follows those found in [2, 22], but with some further justification for a better overall understanding.

We will derive this PDE when we derive the variational equation in Section 4.3.1, but it is also an application of the Feynman-Kac formula, stating that the price of the European call option with strike $K$ and maturity $T$ must satisfy

$$\frac{\partial C}{\partial t} + rS\frac{\partial C}{\partial S} + \kappa(\theta - V)\frac{\partial C}{\partial V} + \frac{1}{2}VS^2\frac{\partial^2 C}{\partial S^2} + \rho\sigma VS\frac{\partial^2 C}{\partial S\partial V} + \frac{1}{2}\sigma^2 V\frac{\partial^2 C}{\partial V^2} - rC = 0 \tag{4.4}$$

with the terminal condition $C(T, S, V) = (S - K)^+$.

We will mimic the form of the Black-Scholes formula to form the Ansatz for the solution

$$C(S, V, t) = SP_1(x, V, t) - Ke^{-r(T-t)}P_2(x, V, t), \qquad x := \ln S \tag{4.5}$$

with terminal conditions $P_j(x, V, T) = \mathbf{1}_{\{x \geq \ln K\}}$ so that our Ansatz reduces to $(S - K)^+$ at maturity. We can substitute the Ansatz into the PDE and equate the coefficients of $S$ and $Ke^{-r(T-t)}$ to obtain two PDEs for $P_1$ and $P_2$:

$$\frac{\partial P_j}{\partial t} + (r + u_j V)\frac{\partial P_j}{\partial x} + (a - b_j V)\frac{\partial P_j}{\partial V} + \frac{1}{2}V\frac{\partial^2 P_j}{\partial x^2} + \rho\sigma V\frac{\partial^2 P_j}{\partial x \partial V} + \frac{1}{2}\sigma^2 V\frac{\partial^2 P_j}{\partial V^2} = 0 \tag{4.6}$$

where $u_1 = \frac{1}{2}$, $u_2 = -\frac{1}{2}$, $a = \kappa\theta$, $b_1 = \kappa - \rho\sigma$ and $b_2 = \kappa$ [2]. Another interpretation is, by Feynman-Kac, that $P_j$ is the probability $\mathbb{Q}_j(x_T \geq \ln K \mid \mathcal{F}_t)$ where $\mathbb{Q}_j$ is a measure with the same dynamics as $\mathbb{Q}$ but with the drift of the log stock price process $x_t = \ln S_t$ changed to $r + u_j V_t$. This is a change of measure that can be done with Girsanov's theorem [9].

With this in mind, let's define the characteristic functions

$$\phi_j(x, V, t; \omega) = \mathbb{E}^{\mathbb{Q}_j}\left[e^{i\omega x_T} \mid \mathcal{F}_t\right] \tag{4.7}$$

which admits the Ansatz

$$\phi_j(\omega) = \exp\left(C_j(\tau, \omega) + D_j(\tau, \omega)V + i\omega x\right), \qquad \tau := T - t \tag{4.8}$$

since the coefficients in the PDE for $P_j$ are independent of $x$ (justifying the $e^{i\omega x}$ term) and affine in $V$ (justifying the $e^{D_j V}$ term)[23].

We can insert this Ansatz into the PDE for $P_j$ to get a system of ODEs for $C_j$ and $D_j$:

$$\frac{\partial D_j}{\partial \tau} = \frac{1}{2}\sigma^2 D_j^2 + (\rho\sigma i\omega - b_j)D_j + u_j i\omega - \frac{1}{2}\omega^2, \qquad \frac{\partial C_j}{\partial \tau} = ri\omega + aD_j \tag{4.9}$$

with the terminal conditions $C_j(0, \omega) = D_j(0, \omega) = 0$. There exists a closed-form solution to this system of ODEs, where we first define

$$d_j := \sqrt{(\rho\sigma i\omega - b_j)^2 - \sigma^2(2u_j i\omega - \omega^2)} \tag{4.10}$$

$$g_j := \frac{b_j - \rho\sigma i\omega - d_j}{b_j - \rho\sigma i\omega + d_j} \tag{4.11}$$

where we use the opposite sign convention compared to Heston's original formulation for better numerical stability (more precisely to avoid regions of numerical discontinuity in the complex plane) as suggested in [24]. This yields the solutions

$$D_j(\tau, \omega) = \frac{b_j - \rho\sigma i\omega - d_j}{\sigma^2} \cdot \frac{1 - e^{-d_j\tau}}{1 - g_j e^{-d_j\tau}} \tag{4.12}$$

$$C_j(\tau, \omega) = ri\omega\tau + \frac{a}{\sigma^2}\left[(b_j - \rho\sigma i\omega - d_j)\tau - 2\ln\frac{1 - g_j e^{-d_j\tau}}{1 - g_j}\right] \tag{4.13}$$

By the Gil-Pelaez inversion theorem [25], we can retrieve $P_j$ from its characteristic function $\phi_j$:

$$P_j = \frac{1}{2} + \frac{1}{\pi}\int_0^\infty \Re\left[\frac{e^{-i\omega\ln K}\phi_j(\omega)}{i\omega}\right]\,\mathrm{d}\omega \tag{4.14}$$

which we can substitute back into our original Ansatz for the call option (4.5) to get the closed-form solution for the call option price.

## 4.2 Tree Solution

Since the closed-form solution is limited to European options, we would like to find ways to approximate solutions to the pricing problem so that we can eventually extend the case to American options, which has no closed-form solution. The first method we will explore is the tree method, where it is easy to modify to the American case by adding an additional comparison to the intrinsic value at each node. We will use the closed-form formula as a benchmark for the tree method for European options, to give us confidence in the solution the tree model provides for American options as it is a simple modification in the algorithm, which we can use as a benchmark for the neural networks we will explore later.

The idea will be similar to the binomial tree we explored earlier in the paper for the Black-Scholes model, but we now have two correlated stochastic processes in the Heston model: the asset price process and the variance process. As this tree is not recombinant, generating paths for these processes and computing the option prices by backwards induction naively would lead to an exponential time complexity, which is not computationally feasible. Therefore, we will use a grid interpolation approach as in [4] so that the number of nodes grows quadratically in the number of time steps instead.

For the algorithm, we will work under the variance process $V$ and log stock price process $Z$ (which is a simple transformation from $S$ using Itô's formula):

$$\mathrm{d}V_t = \kappa(\theta - V_t)\mathrm{d}t + \sigma\sqrt{V_t}\mathrm{d}W_t^1 \tag{4.15}$$

$$\mathrm{d}Z_t = (r - \tfrac{1}{2}V_t)\mathrm{d}t + \sqrt{V_t}\mathrm{d}W_t^2 \tag{4.16}$$

where $\mathrm{d}\langle W^1, W^2\rangle_t = \rho$ under the risk-neutral measure $\mathbb{Q}$ and $\kappa, \sigma, \theta, r > 0$ as before. Of course, the value $C$ of the European option with maturity $T$ and payoff function in terms of stock and volatility values $g : \mathbb{R}^+ \times \mathbb{R}^+ \to \mathbb{R}$ will be

$$C(t, V_t, Z_t) = e^{-r(T-t)}\mathbb{E}^{\mathbb{Q}}[g(e^{Z_t}, \sqrt{V_t}) \mid \mathcal{F}_t] \tag{4.17}$$

### 4.2.1 Euler scheme

To construct the tree, we need to discretise the process. Suppose we have $n$ steps, then $\Delta t = \frac{T}{n}$. Using the standard Euler scheme, we can have

$$V_{k+1} = V_k + \kappa(\theta - V_k^+)\Delta t + Y_{k+1}^1\sigma\sqrt{V_k^+\Delta t} \tag{4.18}$$

$$Z_{k+1} = Z_k + (r - \tfrac{1}{2}V_k)\Delta t + Y_{k+1}^2\sqrt{V_k^+\Delta t} \tag{4.19}$$

where the increment variables $(Y_k^1, Y_k^2)$ are iid in $k$ with probabilities

$$\mathbb{Q}^n(Y_k^1 = i, Y_k^2 = j) = \frac{1}{4}(1 + ij\rho) \qquad i, j \in \{-1, 1\} \tag{4.20}$$

We use a different measure $\mathbb{Q}^n$ as it is not the same as the risk-neutral measure in the Heston model. To ensure positivity of the process, we have taken the positive part of the variance process $V_k^+ := \max(0, V_k)$.

### 4.2.2 Interpolation optimisation

The problem with the naïve Euler scheme is that the tree is not recombining, and hence the number of nodes scales exponentially with increasing time steps, which is not feasible computationally. We will hence, for each time step $k$, construct a grid of option prices at time $k$ and the backward induction would be a bilinear interpolation of the four nearest points.

To set this up, we need the boundaries of the grid at each time step $k$, which is given by the minimum and maximum values of $V_k$ and $Z_k$ in the support. In other words, we have

$$z_k^{\text{max}} = \max\{z : \mathbb{Q}^n(Z_k = z) > 0\} \tag{4.21}$$
$$z_k^{\text{min}} = \min\{z : \mathbb{Q}^n(Z_k = z) > 0\} \tag{4.22}$$
$$v_k^{\text{max}} = \max\{v : \mathbb{Q}^n(V_k = v) > 0\} \tag{4.23}$$
$$v_k^{\text{min}} = \min\{v : \mathbb{Q}^n(V_k = v) > 0\} \tag{4.24}$$
$$\tag{4.25}$$

With these boundaries, we have the grid spacing given our mesh sizes $m_z, m_v \in \mathbb{N}^+$:

$$\Delta z_k = (z_k^{\text{max}} - z_k^{\text{min}})/m_z \tag{4.26}$$
$$\Delta v_k = (v_k^{\text{max}} - v_k^{\text{min}})/m_v \tag{4.27}$$

and the gridpoints:

$$\hat{S}_k = \left\{ \left( v_k^{\text{min}} + i\Delta v_k, z_k^{\text{min}} + j\Delta z_k \right) \,\middle|\, i \in \{0, \dots, m_v\}, j \in \{0, \dots, m_z\} \right\}$$

We will now define the discretised process $(\tilde{V}, \tilde{Z})$ on the grid space $\hat{\mathcal{S}}_k$. This will involve the Euler scheme functions

$$\nu(y, v, z) = v + \kappa(\theta - v^+)\Delta t + y\sigma\sqrt{v^+\Delta t} \tag{4.28}$$
$$\zeta(y, v, z) = z + (r - \tfrac{1}{2}v)\Delta t + y\sqrt{v^+\Delta t} \tag{4.29}$$

Then using these functions, we can retrieve the next step. We can evolve our process to the next step with

$$(\tilde{V}_{k+1}, \tilde{Z}_{k+1}) = \text{snap}^{\hat{\mathcal{S}}_{k+1}}_{Y^3_{k+1}, Y^4_{k+1}} \left( \nu(Y^1_{k+1}, \tilde{V}_k, \tilde{Z}_k), \zeta(Y^1_{k+1}, \tilde{V}_k, \tilde{Z}_k) \right) \tag{4.30}$$

Where $\text{snap}^{\mathcal{S}}_{i_3,i_4} : \mathbb{R}^2 \to \mathcal{S} \subset \mathbb{R}^2$ is a function mapping $(v, z) \in \mathbb{R}^2$ to one of the grid points in $\mathcal{S} := \mathcal{S}_v \times \mathcal{S}_z$. $i_3, i_4 \in \{0, 1\}$ are additional parameters that determine the corner in which $(v, z)$ is mapped. Visually, this is illustrated in Figure 4.1, where each point in the price-variance space can evolve to four different points in the next time step, each of which we take the interpolation of the four nearest grid points.

To do this, we define the 1D snapping functions $\phi_i^{\mathcal{T}} : \mathbb{R} \to \mathcal{T}$ where we have the one-dimensional grid points $\mathcal{T} \subset \mathbb{R}$ and the snap-right indicator $i \in \{0, 1\}$. The function is defined mathematically as

$$\phi_0^{\mathcal{T}}(u) = \max\{u' \in \mathcal{T} : u' \leq u\} \tag{4.31}$$
$$\phi_1^{\mathcal{T}}(u) = \min\{u' \in \mathcal{T} : u' \geq u\} \tag{4.32}$$

Our 2D snapping function $\text{snap}^{\mathcal{S}}_{i_3,i_4} : \mathbb{R}^2 \to \mathcal{S}$ where $\mathcal{S} = \mathcal{S}_v \times \mathcal{S}_z \subset \mathbb{R}^2$ is thus

$$(v, z) \mapsto \left( \phi_{i_3}^{\mathcal{S}_v}(v), \phi_{i_4}^{\mathcal{S}_z}(z) \right) \tag{4.33}$$

The joint probabilities [4] for $Y_{k+1}$ is given by, for $i_1, i_2 \in \{-1, 1\}, i_3, i_4 \in \{0, 1\}$:

$$q_{i_1,i_2,i_3,i_4}(\tilde{V}_k, \tilde{Z}_k) := \tilde{\mathbb{Q}}(Y^1_{k+1} = i_1, Y^2_{k+1} = i_2, Y^3_{k+1} = i_3, Y^4_{k+1} = i_4 \mid (\tilde{V}_k, \tilde{Z}_k)) \tag{4.34}$$
$$= \frac{1}{4}(1 + i_1 i_2 \rho) \times c^{\hat{\mathcal{S}}_{k+1}}_{i_3 i_4} \left( \nu(i_1, \tilde{V}_k, \tilde{Z}_k), \zeta(i_2, \tilde{V}_k, \tilde{Z}_k) \right) \tag{4.35}$$

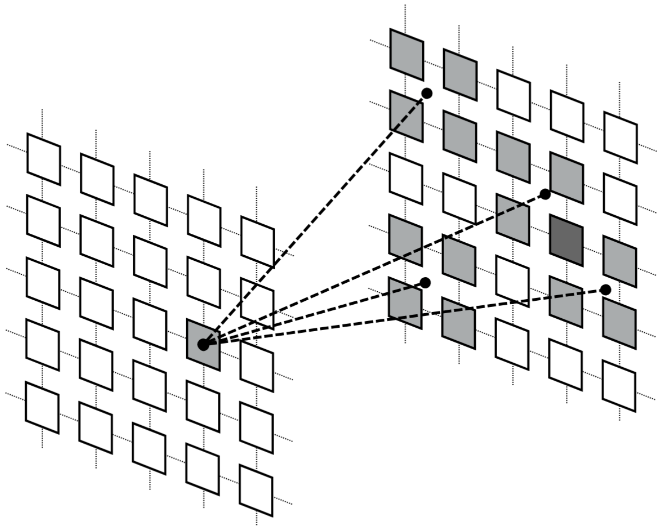

**Figure 4.1** Interpolation illustration (adapted from [4])

Here we have used the interpolation quantity [26], where on a general grid set $\mathcal{S} = S_v \times S_z \subset \mathbb{R}^2$ we have

$$c^{\mathcal{S}}_{i_3 i_4}(v,z) = \begin{cases} (1-\tilde{v})(1-\tilde{z}) & (i_3,i_4) = (0,0) \\ \tilde{v}(1-\tilde{z}) & (i_3,i_4) = (1,0) \\ (1-\tilde{v})\tilde{z} & (i_3,i_4) = (0,1) \\ \tilde{v}\tilde{z} & (i_3,i_4) = (1,1) \end{cases}$$

where we have the difference terms

$$\tilde{v} = \frac{v - \phi_0^{\mathcal{S}_v}(v)}{\phi_1^{\mathcal{S}_v}(v) - \phi_0^{\mathcal{S}_v}(v)} = \frac{v - \phi_0^{\mathcal{S}_v}(v)}{\Delta v_k} \tag{4.36}$$

$$\tilde{z} = \frac{z - \phi_0^{\mathcal{S}_z}(z)}{\phi_1^{\mathcal{S}_z}(z) - \phi_0^{\mathcal{S}_z}(z)} = \frac{z - \phi_0^{\mathcal{S}_z}(z)}{\Delta z_k} \tag{4.37}$$

because our grid is evenly spaced at all time steps $k$.

### 4.2.3 Backwards induction

We can now work backwards on the tree, having constructed the grids. Our base case is the final grid, where the price is just the payoff as we are at maturity. For a general time $t_k$ at step $k$ and node $(v,z) \in \hat{\mathcal{S}}_k$, we will generate the four possible steps determined by $y_1, y_2$ each taking values in $\{-1, 1\}$, and evolve our node with

$$(v,z) \mapsto \big(\nu(y_1,v,z), \zeta(y_2,v,z)\big) =: (v', z') \tag{4.38}$$

Which will be in between four points on the grid $\hat{\mathcal{S}}_{k+1}$. These can be found by performing the snap function on our evolved node: for $y_3, y_4 \in \{0,1\}$ we can encapsulate $y = (y_1, y_2, y_3, y_4)$ and we have

$$(v_y, z_y) = \mathrm{snap}^{\hat{\mathcal{S}}_{k+1}}_{y_3,y_4}(v', z') \tag{4.39}$$

Each of these will have a known option price. Therefore, we can just compute the discounted expectation of these sixteen nodes (two choices for each of $y_1, \ldots, y_4$) for the option price at the original node:

$$C(t_k, v, z) = e^{-r\Delta t} \sum_{y \in \mathcal{Y}} q_{y_1,y_2,y_3,y_4}(v, z) \times C(t_{k+1}, v_y, z_y) \tag{4.40}$$

where $\mathcal{Y} = \{-1, 1\}^2 \times \{0, 1\}^2$.

### 4.2.4 Implementation techniques

If we implemented this method naively, it would be very difficult for us to use the binomial tree as a benchmark because it would need to construct a new tree from scratch for every single value of $S_0$, $V_0$ and $T$. Thus any heatmaps or plots we create would require hundreds of trees which is not feasible, considering each tree requires many time steps and large mesh sizes to be accurate.

Thus in addition to vectorising our code for each tree, we will instead consider an initial range $[S_0^{\text{min}}, S_0^{\text{max}}]$ and $[V_0^{\text{min}}, V_0^{\text{max}}]$ which we will use to construct our grid bounds. More specifically, $z_k^{\text{max}}$ and $v_k^{\text{max}}$ will be the up-most path starting with $(S_0^{\text{max}}, V_0^{\text{max}})$ instead of simply some individual value $(S_0, V_0)$, and similar for the down paths. In this way, when we use the binomial tree as a benchmark, we can use any initial value of $(S_0, V_0)$ as long as it's in the original range and compute the price with an interpolation as described above. With this method, we can evaluate across different initial values while constructing just one tree. We sacrifice a little accuracy in doing so because the grids would be more widely spaced, which we can combat by increasing the mesh sizes. Thus, we effectively construct one more complex tree instead of hundreds of simpler ones, which is much more computationally efficient.

As for selecting our mesh sizes, we will use the result that if we have

$$\lim_{n\to\infty} \Delta t = 0, \qquad \lim_{n\to\infty} \max_{k=1,\dots,n-1} \frac{\Delta v_k^n}{\Delta t_k} = 0, \qquad \lim_{n\to\infty} \max_{k=1,\dots,n-1} \frac{\Delta z_k^n}{\Delta t_k} = 0 \tag{4.41}$$

then the process represented in our Heston tree converges weakly to the original process $(V, Z)$ [4]. Note here we denote $\Delta v_k$ from before as $\Delta v_k^n$ to emphasize we have split our maturity $T$ into $n$ steps, and similarly for $\Delta z_k^n$.

Bearing this in mind, a resolution of $n = 100, m_v = 300, m_z = 600$ allows our tree to faithfully reproduce the closed-form solution, which are presented in the numerical results in section 4.5.1.

## 4.3 PINN Solution

### 4.3.1 Variational equation

We would now like to derive the variational equation for options on assets under the Heston model. Note that for a general process $X_t \in \mathbb{R}^d$ solving the Itô SDE

$$\mathrm{d}X_t = \boldsymbol{b}(X_t)\mathrm{d}t + \boldsymbol{\sigma}(X_t)\mathrm{d}\boldsymbol{W}_t \tag{4.42}$$

where we have the drift $\boldsymbol{b} : \mathbb{R}^d \to \mathbb{R}^d$, the volatility $\boldsymbol{\sigma} : \mathbb{R}^d \to \mathbb{R}^{d\times m}$ and the $m$-dimensional Brownian motion $\boldsymbol{W}_t$. On second-order smooth functions $f \in C^2(\mathbb{R}^d)$, the infinitesimal generator $\mathcal{L}$ is defined as [27]

$$\mathcal{L}f(x) := \lim_{t\to 0} \frac{\mathbb{E}[f(X_t) \mid X_0 = x] - f(x)}{t} \tag{4.43}$$

which exists and can be derived with Itô's formula (or Taylor's expansion) to be [28]

$$\mathcal{L}f(x) = \sum_{i=1}^{d} \boldsymbol{b}_i(x) \frac{\partial f}{\partial x_i} + \frac{1}{2} \sum_{i=1}^{d} \sum_{j=1}^{d} \left(\boldsymbol{\sigma}\boldsymbol{\sigma}^\top\right)_{ij}(x) \frac{\partial^2 f}{\partial x_i \partial x_j} \tag{4.44}$$

Let's apply this to our Heston model. The drift vector is

$$\boldsymbol{b}(S,V) = \begin{pmatrix} rS \\ \kappa(\theta - V) \end{pmatrix} \tag{4.45}$$

To compute $\boldsymbol{\sigma}\boldsymbol{\sigma}^\top$, we need to represent our correlated Brownian motions in terms of independent ones. Suppose we have independent Brownian motions $B^1$ and $B^2$, we can define

$$W^1 = B^1 \tag{4.46}$$

$$W^2 = \rho B^1 + \sqrt{1-\rho^2} B^2 \tag{4.47}$$

which has the quadratic variation $\mathrm{d}\langle W^1, W^2 \rangle_t = \rho \mathrm{d}t$ as required. Using these new Brownian motions, we can rewrite the SDE as

$$\mathrm{d}\begin{pmatrix} S_t \\ V_t \end{pmatrix} = \underbrace{\begin{pmatrix} rS_t \\ \kappa(\theta - V_t) \end{pmatrix}}_{\boldsymbol{b}(S_t,V_t)} \mathrm{d}t + \underbrace{\begin{pmatrix} S_t\sqrt{V_t} & 0 \\ \rho\sigma\sqrt{V_t} & \sigma\sqrt{V_t}\sqrt{1-\rho^2} \end{pmatrix}}_{\boldsymbol{\sigma}(S_t,V_t)} \begin{pmatrix} \mathrm{d}B_t^1 \\ \mathrm{d}B_t^2 \end{pmatrix} \tag{4.48}$$

which means that our covariance matrix is now

$$\boldsymbol{\sigma}(S,V) = \begin{pmatrix} S\sqrt{V} & 0 \\ \rho\sigma\sqrt{V} & \sigma\sqrt{V}\sqrt{1-\rho^2} \end{pmatrix}. \tag{4.49}$$

which means

$$\boldsymbol{\sigma}\boldsymbol{\sigma}^\top = \begin{pmatrix} S^2V & \rho\sigma SV \\ \rho\sigma SV & \sigma^2 V \end{pmatrix} \tag{4.50}$$

We now have everything we need for the formula. For any function $f = f(S,V) \in C^2(\mathbb{R}^d)$, we have

$$\mathcal{L}f = rS\frac{\partial f}{\partial S} + \kappa(\theta - V_t)\frac{\partial f}{\partial V} + \frac{1}{2}\left(S^2V\frac{\partial^2 f}{\partial S^2} + 2\rho\sigma SV\frac{\partial^2 f}{\partial S \partial V} + \sigma^2 V\frac{\partial^2 f}{\partial V^2}\right) \tag{4.51}$$

This is our infinitesimal generator $\mathcal{L}$ acting on smooth functions $f$. Suppose we have a European option with payoff $g: \mathbb{R}^+ \to \mathbb{R}^+$, then our price function $u(t,S,V)$, the value of the call option with expiry $T$ at time $t$ with $S_t = S$ and $V_t = V$ satisfies

$$\mathcal{G}[u] := ru - \frac{\partial u}{\partial t} + \mathcal{L}u = 0 \tag{4.52}$$

with terminal condition $u(T,S,V) = g(S)$. This matches the PDE we had stated in the closed-form solution in section 4.4.

As a check, we can set $\kappa = 0$ and $\sigma = 0$ so that $\mathrm{d}V = 0$ and volatility becomes a constant, and our model collapses to what we have been working with before. Our generator becomes

$$\mathcal{L}f = rS\frac{\partial f}{\partial S} + \frac{1}{2}S^2 V_0 \frac{\partial^2 f}{\partial S^2} \tag{4.53}$$

which matches our expression (3.35) from the Black-Scholes section, since $V_0$ is the initial (and hence constant) variance.

We have derived the variational equation for the European option for simplicity, but it can be easily extended for the American case. With the same argument as (3.37), the value of the American option must satisfy the PDE

$$\min\left(\mathcal{G}[u], v(t,S,V) - g(S)\right) = 0 \tag{4.54}$$

### 4.3.2 PINN formulation

We can fit the variational equation (4.54) into the PINN framework established in section 2.3 which will serve as the PDE loss (2.36), and all that is left is to set up the loss function and the relevant boundary conditions. It will be very similar to the Black-Scholes case, but as we have an additional dimension $V$ for the variance, we will need additional boundary conditions to ensure regularity in the variance dimension as well. These boundary conditions will be inspired by those given for the European call option in [2], but we will adapt them in our study, as they exhibit different behaviour for American put options.

1. $t \to T$: the value at expiry is just the intrinsic value, or the payoff of the put option:

$$v(T, S, V) = \max(0, K - S) \tag{4.55}$$

which will form the terminal loss (2.37) for the PINN.

2. $S \to 0$: the option value is the strike when the stock is worthless, because we can exercise immediately (for European options we would require a discounting term):

$$v(t, 0, V) = K \tag{4.56}$$

3. $S \to \infty$: the value of a very far out of the money option is 0, as it is very unlikely for it to expire in of the money:

$$v(t, \infty, V) = 0 \tag{4.57}$$

4. $V \to 0$: volatility being zero is a slightly more complicated case as Heston proposes a collapsed first-order PDE, which does not yield good results in practice. Instead, we will set the condition to be the payoff under the guise that asset price becomes deterministic, which is only true instantaneously due to the mean-reverting nature of the variance process, but it is a good approximation for small volatilities:

$$v(t, S, 0) = \max(0, K - S) \tag{4.58}$$

5. $V \to \infty$: when the volatility is very high, it is almost sure that the product of assets would be zero at some point before expiry, so the value of the option is the strike:

$$v(t, S, \infty) = K \tag{4.59}$$

In practice, as we clearly cannot represent infinity, we will settle for large constants for the asset price $S_{\text{inf}}$ and variance $V_{\text{inf}}$. The minimum for both of these will be taken to be zero. Thus our domain is

$$\Omega = (0, T) \times (0, S_{\text{max}}) \times (0, V_{\text{max}}) \tag{4.60}$$

The boundary conditions (2)-(5) together will form the boundary loss (2.38) for the PINN. By choosing relevant weights for each of these loss components (we could additionally set different values for different individual boundary components), we can ensure that the neural network learns the correct behaviour at the boundaries, which is crucial for the accuracy of the solution.

## 4.4 Multi-asset PINN

With the neural network established, we are now ready to move on to pricing options on multiple assets. As with the Black-Scholes case, we will price the put on the product of multiple assets because we can find a one-dimensional reduction. Here, we will consider a simplified model. Suppose we have $n$ assets and our assets $S = (S_1, \dots, S_n)^\top$ are governed by the stochastic differential equations:

$$\mathrm{d}S = \text{diag}(S)(r\,\mathrm{d}t + \sqrt{V}\sigma\Sigma\,\mathrm{d}W_1) \tag{4.61}$$

$$\mathrm{d}V = \kappa(\theta - V)\,\mathrm{d}t + \bar{\sigma}\sqrt{V}\,\mathrm{d}W_2 \tag{4.62}$$

where we have the vector of iid Brownian motions $W_1 = (W_{1,1}, \dots, W_{1,n})^\top$, variance vector $\sigma = (\sigma_1, \dots, \sigma_n)^\top$ and the covariance matrix $\Sigma$, where for simplicity we will assume all cross-correlations $\Sigma_{ij} = \rho$ for $i \neq j$. Assume additionally that each asset is correlated with the variance process: $\mathrm{d}W_{1,j}\mathrm{d}W_2 = \rho_j \mathrm{d}t$. Note that this means the asset $S_i$ satisfies

$$\mathrm{d}S_i = S_i \left( r\,\mathrm{d}t + \sqrt{V}\sigma_i \sum_{j=1}^{n} \Sigma_{ij}\,\mathrm{d}W_{1,j} \right) \tag{4.63}$$

To guarantee positiveness of the variance process, we will enforce the Feller condition on the parameters

$$2\kappa\theta \geq \bar{\sigma}^2 \tag{4.64}$$

We make these assumptions to allow a one-dimensional reduction for the Heston model, which would provide us a comparable benchmark for our results with the tree method. This would in turn give us confidence for our approximated solutions, should we wish to relax assumptions or to change the payoff.

### 4.4.1 One-dimensional reduction

Since we would like to find an equivalent formulation of the product of assets $X := \prod_{i=1}^{n} S_i$, it would be easier to work in the log domain. That is, find the governing SDE for the log process $Y = \log X = \sum_{i=1}^{n} \log S_i$ and then exponentiate with Itô's lemma to recover the desired result.

Let $Y_i = \log S_i$. Then by Itô's lemma, we have

$$\mathrm{d}Y_i = \frac{1}{S_i}\,\mathrm{d}S_i - \frac{1}{2S_i}\,\mathrm{d}S_i\,\mathrm{d}S_i \tag{4.65}$$

$$= r\,\mathrm{d}t + \sqrt{V}\sigma_i \sum_{j=1}^{n} \Sigma_{ij}\,\mathrm{d}W_{1,j} - \frac{1}{2}V\sigma_i^2\,\mathrm{d}t \tag{4.66}$$

because cross-correlations are zero within $W_1$. We can collect terms to get

$$\mathrm{d}Y = \left( nr - \frac{1}{2}V \sum_{i=1}^{n} \sigma_i^2 \right)\,\mathrm{d}t + \sqrt{V} \sum_{j=1}^{n} \sum_{i=1}^{n} \sigma_i \Sigma_{ij}\,\mathrm{d}W_{1,j} \tag{4.67}$$

For ease of notation, let $A = \sum_{i=1}^{n} \sigma_i^2$ and $b_j = \sum_{i=1}^{n} \sigma_i \Sigma_{ij}$. Then we have

$$\mathrm{d}Y = \left( nr - \frac{1}{2}VA \right)\,\mathrm{d}t + \sqrt{V} \sum_{j=1}^{n} b_j\,\mathrm{d}W_{1,j} \tag{4.68}$$

Now $Y$ has variance

$$\mathrm{d}Y\,\mathrm{d}Y = V\sum_{j=1}^{n} b_j^2\,\mathrm{d}t =: VB\,\mathrm{d}t \tag{4.69}$$

where we let $B := \sum_{j=1}^{n} b_j^2$ for convenience, we can rewrite $\mathrm{d}Y$ with a new Brownian motion:

$$\mathrm{d}Y = \left(nr - \frac{1}{2}VA\right)\,\mathrm{d}t + \sqrt{VB}\,\mathrm{d}W_X \tag{4.70}$$

where we have

$$\mathrm{d}W_X = \frac{\sum_{j=1}^{n} b_j\,\mathrm{d}W_{1,j}}{\sqrt{B}} \tag{4.71}$$

Using our correlation structure, we have

$$b_j = \sigma_j + \rho\sum_{i\neq j}\sigma_i = (1-\rho)\sigma_j + \rho\underbrace{\sum_{i=1}^{n}\sigma_i}_{=:S_\sigma} \tag{4.72}$$

So the sum of squares $B$ is just

$$B = \sum_{j=1}^{n} b_j^2 = \sum_{j=1}^{n}\left((1-\rho)\sigma_j + \rho S_\sigma\right)^2 \tag{4.73}$$

$$= (1-\rho)^2 A + S_\sigma^2(2\rho(1-\rho) + n\rho^2) \tag{4.74}$$

We now have all components to exponentiate to recover $X = e^Y$. Using Itô's lemma again, we have

$$\mathrm{d}X = X\,\mathrm{d}Y + \frac{1}{2}X\,\mathrm{d}Y\,\mathrm{d}Y \tag{4.75}$$

$$= X\left(nr + \frac{1}{2}V(B-A)\right)\,\mathrm{d}t + X\sqrt{VB}\,\mathrm{d}W_X \tag{4.76}$$

This is close to the Heston model, with the diffusion $\tilde{V} = VB$ which gives us

$$\mathrm{d}\tilde{V} = B\,\mathrm{d}V \tag{4.77}$$

$$= \kappa(B\theta - \tilde{V})\,\mathrm{d}t + \bar{\sigma}\sqrt{B}\sqrt{\tilde{V}}\,\mathrm{d}W_2 \tag{4.78}$$

which is a new CIR process $\tilde{V}$ with parameters $\tilde{\kappa} = \kappa$, $\tilde{\theta} = B\theta$ and $\tilde{\sigma} = \bar{\sigma}\sqrt{B}$. The product of assets nearly gives a direct reduction to a one-dimensional Heston process, but the drift of $X$ contains $V$ which is a stochastic process itself. Thus for simplicity and for our benchmark, we will set $\rho = 0$ which means $B = A$ and the drift term becomes deterministic. As such, we are able to recover a one-dimensional Heston model for the product of assets $X$ with variance $\tilde{V}$ and drift $nr$.

### 4.4.2 Variational equation

Recall for a process $X_t \in \mathbb{R}^d$ solving $\mathrm{d}X_t = b(X_t)\mathrm{d}t + \sigma(X_t)\mathrm{d}W_t$, the infinitesimal generator $\mathcal{L}$ can be derived with Itô's formula [28] on $f(x)$ to be

$$\mathcal{L}f(x) = \sum_{i=1}^{d} b_i(x)\frac{\partial f}{\partial x_i} + \frac{1}{2}\sum_{i=1}^{d}\sum_{j=1}^{d}\left(\sigma\sigma^\top\right)_{ij}(x)\frac{\partial^2 f}{\partial x_i \partial x_j} \tag{4.79}$$

In our case, we have a function $f = f(S_1, \dots, S_d, V)$, and we can derive the infinitesimal generator much like the 1D case, but this time we need to sum over our different asset and variance processes. Mathematically, our generator can be expressed to be

$$\mathcal{L}f = \sum_{i=1}^{d} rS_i \frac{\partial f}{\partial S_i} + \kappa(\theta - V)\frac{\partial f}{\partial V} + \frac{1}{2}\sum_{i=1}^{d}\sum_{j=1}^{d} \Sigma_{ij} V S_i S_j \frac{\partial^2 f}{\partial S_i \partial S_j} \tag{4.80}$$

$$+ \frac{1}{2}\bar{\sigma}^2 V \frac{\partial^2 f}{\partial V^2} + \sum_{i=1}^{d} \rho_i \sigma_i \bar{\sigma} V_i S_i \frac{\partial^2 f}{\partial S_i \partial V} \tag{4.81}$$

For any payoff $g(\cdot)$, we have, as before, the variational equation for an European option being

$$\mathcal{G}[f] := rf - \frac{\partial f}{\partial t} + \mathcal{L}f = 0 \tag{4.82}$$

and for an American option, we can introduce the early stopping criterion and the variation equation becomes

$$\min(\mathcal{G}[f], f - g) = 0 \tag{4.83}$$

### 4.4.3 PINN formulation

This will be very similar to the one-dimensional case, but we will make a few modifications to account for the additional dimensions. The PDE loss will be the same as before, but the boundary conditions will include:

1. $t \to T$: the value at expiry is just the intrinsic value which is the payoff:

$$u(T, S_1, \dots, S_n, V) = \max\left(0, K - \prod_{k=1}^{n} S_k\right) \tag{4.84}$$

   which serves as our terminal loss (2.37) for the PINN.

2. $S \to 0$: if one of the assets reaches zero, then the product of assets is consequently zero and we should exercise immediately. Thus the value is the strike. To implement this, let $\tilde{\boldsymbol{S}}$ be $\boldsymbol{S}$ with one of the assets modified manually to $0$. Then we have

$$v(t, \tilde{\boldsymbol{S}}, V) = K \tag{4.85}$$

3. $S \to \infty$: if one of the assets has a very high price, then the product of assets would also be very high meaning the option would expire worthless out of the money. To implement this, let $\bar{\boldsymbol{S}}$ be $\boldsymbol{S}$ with one of the assets modified manually to $S_{\text{inf}}$, a large constant. Then the boundary condition is

$$v(t, \bar{\boldsymbol{S}}, V) = 0 \tag{4.86}$$

4. $V \to 0$: with the same logic as the one-dimensional case, we will set the price under zero volatility to be zero. Hence we have

$$v(t, \boldsymbol{S}, 0) = \max\left(0, K - \prod_{k=1}^{n} S_k\right) \tag{4.87}$$

5. $V \to \infty$: with infinite volatility, it is almost sure that the intrinsic value of the option would reach the strike at some time before expiry, and thus the value of the option is the strike for the put. To implement this, set $V_{\text{inf}}$ a large constant, and we have

$$v(t, \boldsymbol{S}, V_{\text{inf}}) = K \tag{4.88}$$

## 4.5 Numerical Results

### 4.5.1 Interpolating tree method

As mentioned before, we choose the resolution $n = 100, m_v = 300, m_z = 600$ which allows a faithful approximation of closed form solution while keeping runtime low. Figure 4.2 are the results after using the parameters $K = 1, T = 1, r = 0.05, \kappa = 2, \theta = 0.04, \sigma = 0.3, \rho = -0.7$. We produce these plots by varying across different values of $S_0$ (for delta), $v_0$ (for vega) and time elapsed (for term structure), and all else kept constant.


**Figure 4.2** Call option prices by the Heston Tree

We can see that the Heston tree does indeed replicate the closed form solution very closely to very small error. This gives us confidence in our implementation, and with an easy modification we can compute the American option prices which we will use as a benchmark.

To compute the American option price with the tree model, all we need to do is to introduce the stopping condition at every node: we evaluate the intrinsic value of the option at each node and set the option value to be the maximum of the intrinsic value and the discounted continuation value.


**Figure 4.3** American vs European option prices for the put

In Figure 4.3 we can see that the European option price indeed bounds the American prices from below, as we would expect. We have confidence that our American prices are accurate since our binomial tree faithfully approximates the closed-form solution for European options. We will now use this as a benchmark for neural network approximations.

### 4.5.2 One-dimensional PINN solution

We are now ready to train the neural network to price our option. We choose to use a learning rate of $1 \times 10^{-3}$ with the Adam optimiser and a scheduler with step size 2000 and gamma 0.7 and the Sigmoid activation function. By using parameters $K = 1, T = 1, r = 0.1, \kappa = 2, \theta = 0.04, \sigma = 0.3, \rho = -0.7$ obtain the plots as shown in Figure 4.4 for the American put.

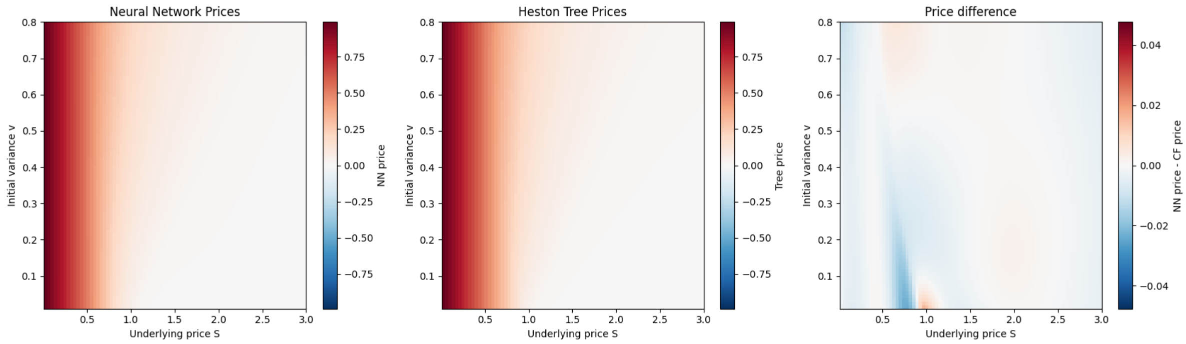


**Figure 4.4** American put option prices

We can see that the neural network indeed faithfully reproduces the solution approximated by the binomial tree, even though it tends to perform less well at the money especially at greater volatilities. This can also be easily shown by fixing a volatility and plotting the approximated option value against the underlying, as shown in in Figure 4.5.

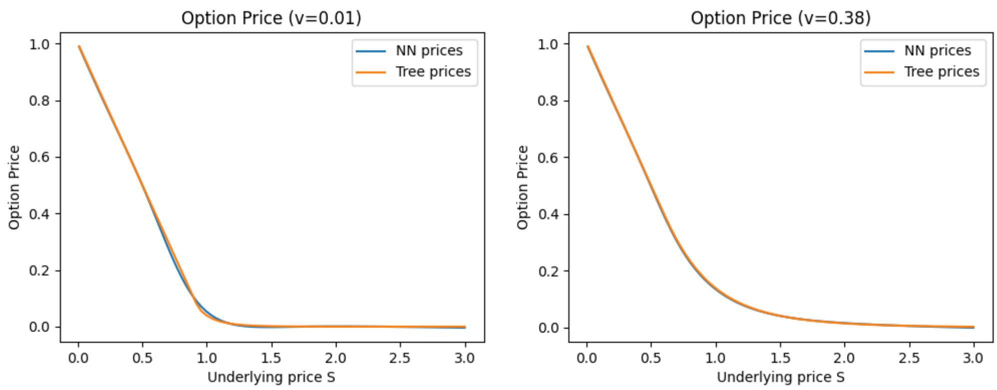


**Figure 4.5** American put option prices with volatility fixed

We can see that the tree prices are indeed well approximated by the Neural network, but the at-the-money options are underpriced for higher volatilities. This is due to the lack of a strict high initial variance boundary condition. We can plot the option prices with respect to initial variance in Figure 4.6. Our price does indeed increase with increasing initial variance, but not at the same rate.

### 4.5.3 Multi-asset PINN solution


**Figure 4.6** At-the-money price with varying initial variance

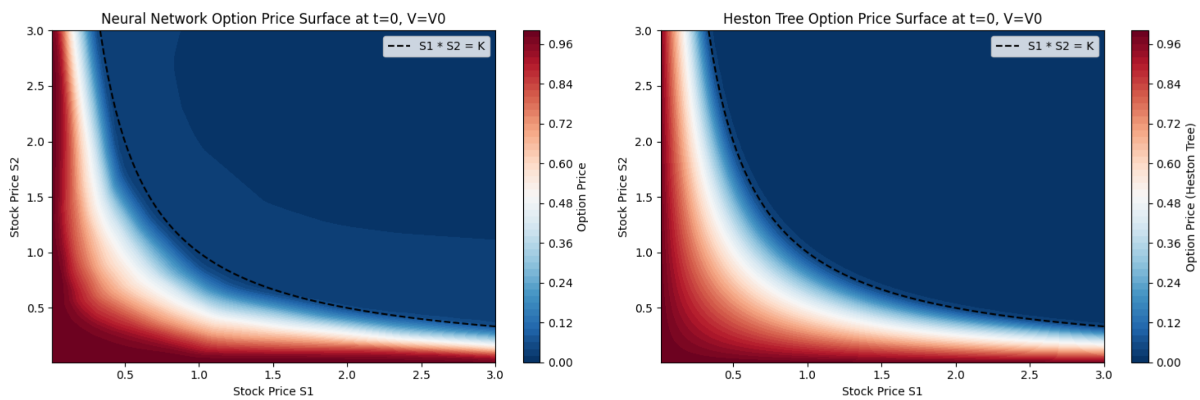


**Figure 4.7** Neural network vs 1D reduction tree prices

# Appendix A

# Appendix 1

## A.1 Change of measure for the Black-Scholes model

Consider the discounted stock price

$$\tilde{S}_t := \frac{S_t}{B_t} = e^{-rt} S_t \tag{A.1}$$

and apply Itô's lemma to get a new SDE in terms of the discounted stock price

$$\mathrm{d}\tilde{S}_t = \tilde{S}_t[(\mu - r)\mathrm{d}t + \sigma \mathrm{d}W_t^{\mathbb{P}}] \tag{A.2}$$

which means that under the physical measure $\mathbb{P}$, $\tilde{S}_t$ has nonzero drift unless $\mu = r$, and hence not a martingale. So to enforce no arbitrage in our pricing model, we need a new equivalent probability measure, name it $\mathbb{Q}$, under which $\tilde{S}_t$ is a martingale.

Let's define the quantity

$$c := \frac{\mu - r}{\sigma} \tag{A.3}$$

which measures how much excess drift per unit volatility the stock earns under the physical measure $\mathbb{P}$. Then we can define the Radon-Nikodym derivative

$$\left.\frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}\right|_{\mathcal{F}_t} = Z_t := \exp\left(-cW_t^{\mathbb{P}} - \frac{1}{2}c^2 t\right) \tag{A.4}$$

which satisfies Novikov's condition as $c$ is constant so $\mathbb{Q}$ is well defined. The new process

$$W_t^{\mathbb{Q}} := W_t^{\mathbb{P}} + ct \tag{A.5}$$

is a Brownian motion under $\mathbb{Q}$ by Girsanov's theorem. If we substitute this into the original SDE, we obtain

$$\frac{\mathrm{d}S_t}{S_t} = \mu \mathrm{d}t + \sigma(W_t^{\mathbb{Q}} - c\mathrm{d}t) \tag{A.6}$$

$$= (\mu - \sigma c)\mathrm{d}t + \sigma \mathrm{d}W_t^{\mathbb{Q}} \tag{A.7}$$

$$= r\mathrm{d}t + \sigma \mathrm{d}W_t^{\mathbb{Q}} \tag{A.8}$$

So now under $\mathbb{Q}$, we have $\mathrm{d}\tilde{S}_t = \sigma \tilde{S}_t \mathrm{d}W_t^{\mathbb{Q}}$ which has zero drift, and hence $\tilde{S}_t$ is a $\mathbb{Q}$-martingale.

Intuitively, this change of measure reweights possible stock paths so that there is no more risk premium, and all assets earn the risk-free rate in expectation after discounting.

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